

Review article

Interface engineering in conducting polymers-based supercapacitor

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ABSTRACT

Conducting polymers (CPs), such as polypyrrole, polyaniline, polythiophene, poly (3,4-ethylenedioxythiophene) and polyindole are currently the focus of extensive research in supercapacitors due to their conjugated structures, variable oxidation states, and efficient doping and de-doping processes, which enhance areal/specific capacitance. However, pure CP electrodes encounter challenges such as lower energy density and inadequate cycling stability, hampering their widespread industrial application. Interface interactions between different materials can alter the electronic states and physicochemical properties of composites, thereby significantly impacting electrochemical performance. In this review, we explore interface engineering strategies employed in the fabrication of novel CPs-based composite electrode materials, which involve modifying CPs with carbon-based materials, transition metal oxides, metal organic frameworks and other commonly used nanomaterials. These strategies include hydrogen bonding, π - π interactions, electrostatic interactions, confinement effects, and synergistic effects. We emphasize the mechanisms underlying the enhancement of CPs-based composites and detailed various preparation methods. By providing insights into the preparation and utilization of CPs-based composites for supercapacitors, this review contributes valuable perspectives for future research directions.

1. Introduction

In recent years, supercapacitors (SCs) have developed rapidly in the field of energy storage, owing to their high specific power density, inherent stability, and wide operating temperature range [1–3]. The applications in automobile starters, electromagnetic ejection systems, and more are widespread [4]. However, their relatively low specific energy density compared to secondary batteries hampers further commercialization. The discovery of pseudocapacitive materials such as metal oxides, metal carbides, and conducting polymers (CPs) with surface redox processes has led to a 10 to 100-fold increase in interfacial capacitance initiating intense research for materials and modifications [5].

Electronically CPs possess high conductivity, lower band-gap (1–3 eV), suitable morphology, ease of synthesis and high intrinsic flexibility [6]. However, two major drawbacks have hindered their widespread success: significant degradation when exposed to electrolyte solutions and cycled between various states of oxidation during charge and discharge, and decreased electronic conductivity due to structural and molecular changes [7,8]. These detrimental effects, such as decreasing conductance or low energy density, are often mitigated by incorporating

conducting carbon-based materials or high-capacitance materials, respectively [9]. Consequently, CPs-based composites, which combine metal oxides or carbon-based materials, hold promise as solutions to these challenges.

Typically, carbonaceous materials (such as carbon nanotubes, graphene, etc.), metal oxides, or a combination of both are integrated with CPs to enhance specific capacitance and longevity. However, mere mixing often results in electrode component aggregation, thereby exacerbating SCs performance [7]. Hence, a universal and straightforward approach, such as interface engineering, is essential to optimize the dispersion of the building blocks, electrical conductivity, number of active sites and charge transfer kinetics to improve specific capacitance and cycle stability in CPs-based composites. Electronic couplings, including hydrogen bonds, π - π interactions, electrostatic interactions, and synergistic effects at the interface, can significantly enhance the SC performance of CPs-based systems [10,11]. For instance, the synergy property increases the charge transfer efficiency of composites and performances [12]. Moreover, the potential interaction between the polymer and additives increases the number of active sites for electron migration, thus preventing polymer degradation and resulting in high energy density and extended lifetime [13,14].

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At present, several reviews exist on CPs-based SCs electrode materials, primarily focusing on recent research involving polyaniline (PANI), polypyrrole (PPy), derivative of polythiophene (PTh) and polyindole (PIn), along with their binary and ternary composites. These reviews delve into various aspects such as synthesis methods, applications, and enhanced electrochemical properties, etc., [7,15–19]. However, a critical discussion of interface effect caused by the interface contact such as effect of noncovalent interaction, confinement effect and synergetic effect in the CPs-based composites for SCs is still rare. Interfacial interactions can yield several positive influences, including enhancing the dispersion of building blocks within the composite, thereby boosting the specific capacitance of double-layer and pseudo-capacitive materials [20]. Simultaneously, this enhancement may mitigate the volume expansion of polymer chains during charge-discharge processes, thereby improving cycling performance. Therefore, in this review, we will initially concentrate on interface engineering methods employed in CPs-based composites, such as CPs/carbon binary composites and CPs/carbon/metal oxide ternary composites. Then, the readers could have a snapshot of the common interface interactions existed in the CPs-based composites for SCs, and concurrently obtain a deep knowledge of current challenges and future trends in this field (Fig. 1). Finally, future outlooks and the existing challenges about the interface engineering towards CPs-based SCs will be discussed.

2. Interface engineering of CPs-based composites

Based on the storage mechanism, SCs can be classified as electrochemical double-layer capacitors (EDLCs) and pseudo-SCs, while sharing common traits such as high-power density, fast charging and discharging rates, and long cycle life [21,22]. EDLCs harness energy by physically separating charge at the interface of a porous carbon electrode and an electrolyte solution. When voltage is applied, ions from the electrolyte gather into double layers of charge at the electrode-electrolyte interface, creating capacitance without engaging in faradaic reactions and thereby maintaining chemical stability. The capacitance of EDLCs is contingent upon factors such as the surface area and pore configuration of the carbon electrodes, alongside the composition of the electrolyte [23]. Pseudo-SCs, offer a versatile energy storage solution, which harness energy through a faradaic process [24]. The faradaic electrode, comprising a metal oxide or CPs, facilitates reversible redox reactions, thereby enhancing energy density beyond that of EDLCs. The overall efficacy of pseudo-SCs hinges directly on the selection of materials for both electrodes and electrolytes [20].

As shown, CPs are attractive due to their high charge density and the possibility of low resistance, high energy, and power density [25]. While, the capacitance of the CPs could not be displayed fully owing to the change in reaction condition, oxidants, electrolytes [26]. Usually,

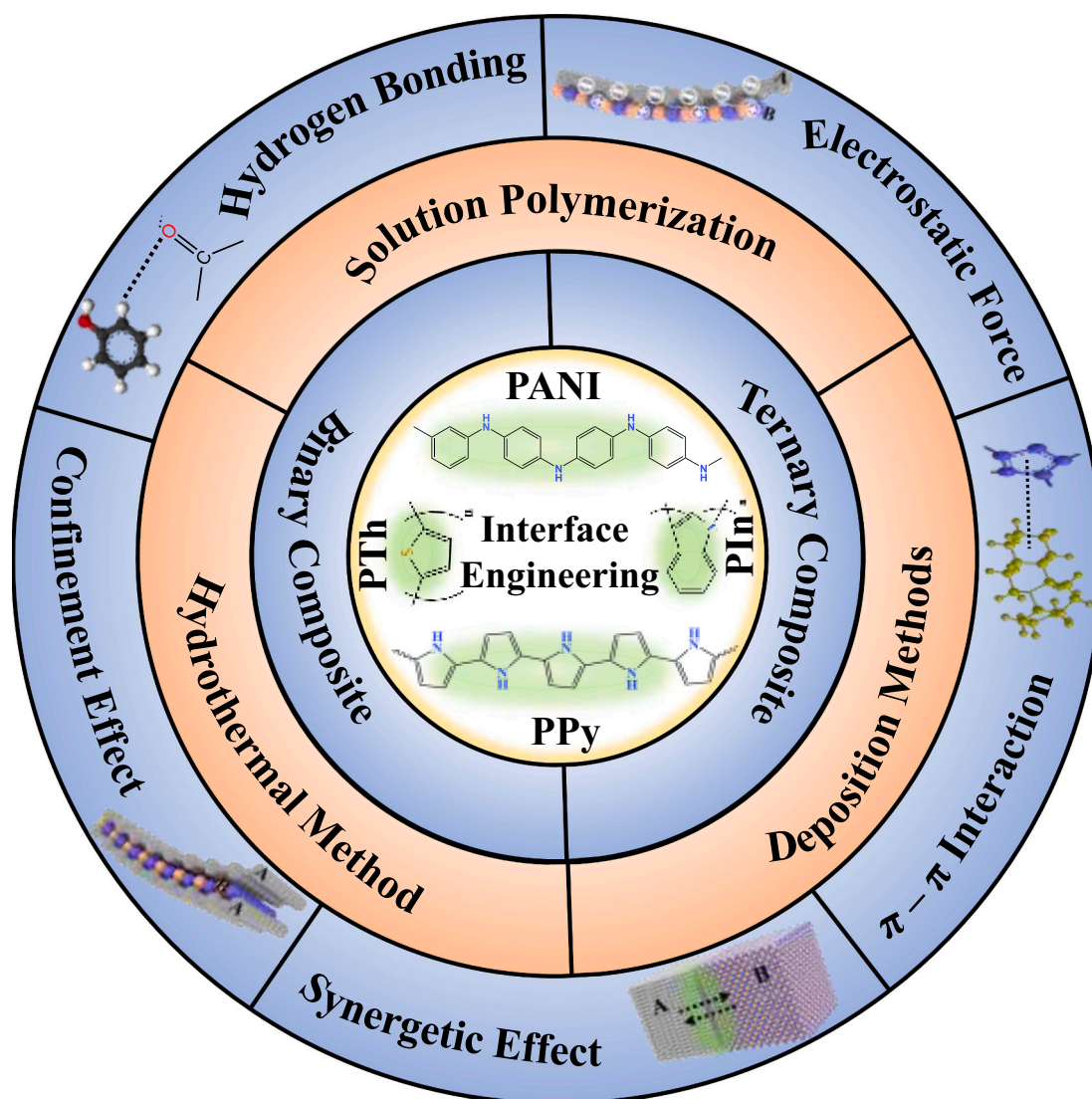


Fig. 1. Illustration of interface engineering of CP-based composites with the synthesis methods and interface effects.

PANI possesses a practical achievable capacitance range between 250 and 950 F/g [27], which is less than the theoretical capacitance 1284 F/g [28], and it exhibits low cycle stability in the neutral electrolyte. The HCl was used as a protonic acid dopant assurance for enhanced conductivity and helped electron transfer in the electrode. Numerous strategies have been proposed to augment the performance of PANI, encompassing the development of composite electrode materials and manipulating electrochemical processes during cycling through structural and morphological adjustments. In addition, PPy possess a theoretical capacitance of 620 F/g, and shows higher degree of flexibility, greater density, and while a capacitance range of only 100–400 F/g can be achieved practically [29]. Despite the observed enhancements in capacitance and energy density, the process of crosslinking in PPy can compromise the inherent structural integrity of the original polymer, potentially leading to degradation in the stability of the electrode material over time. Moreover, PTh exhibits theoretical capacitance of 485 F/g and a capacitance of 134–300 F/g in research, with poor cycle life and low conductivity. In usual, substituting phenyl, ethyl, and alkoxy groups at the 3rd position of the thiophene ring can also improve stability to oxygen and water [30]. Finally, PIn can show a practical achievable capacitance range of 90–120 F/g. Nevertheless, its utility in SCs is constrained by low cyclic stability attributable to polymer chain degradation during prolonged ion exchange in electrochemical reactions. Doping PIn with additional ions or species has been shown to significantly boost conductivity and stability [31].

As discussed, pure CPs display multiple limitations when used solely in SCs. Carbon nanomaterials, known for their high conductivity, are commonly used in EDLCs, while transition metal compounds demonstrate high pseudocapacitance, resulting in good cycle stability and high energy density [32]. To address the challenges of charge transfer acceleration and volume change improvement in CPs chains during charge-discharge cycles, interface engineering strategies are fundamentally applied by combining CPs with the aforementioned active materials such as carbon nanomaterials and transition metal compounds. Interface engineering involves designing and optimizing interface interactions, structures, and compositions to achieve the required morphology, active sites, or charge transfer capabilities necessary for SCs applications. For instance, differences in geometric structure and electronic affinity can initiate charge transfer through interfacial chemical bonding [20,33].

Notably, interfaces of composites for SCs can be classified based on morphological structures into two representative configuration models: supported structure and heterogeneous structure. The supported structure involves loading an active species with a smaller size onto the surface of larger-sized species, such as zero-dimensional (0D) nanoparticles on two-dimensional (2D) nanosheets [34], 0D nanoparticles on three-dimensional (3D) blocks [35], 2D nanosheets on 3D current collectors [36]. A heterogeneous structure is formed by multi-building blocks with comparable sizes that are tightly connected on a specific face, such as Ag/PANI/graphitic carbon nitride ($g\text{-C}_3\text{N}_4$) [37], PPy/carbon nanotubes (CNTs) [38], poly(3,4-ethylenedioxythiophene):poly(styrene sulfonate) (PEDOT:PSS)/molybdenum sulfide (MoS_2) [39], PIn/cobalt oxide (Co_3O_4) [40] laying the foundation for constructing unique heterostructure materials.

Nanostructured additives typically possess a significantly larger interface area compared to micron or macro-sized additives by orders of magnitude [41]. Therefore, achieving uniform dispersion of nano-additives within the polymer framework becomes critically important, requiring effective synthesis techniques. Additionally, alongside interface construction, accurately characterizing the physicochemical properties of interfaces in CPs-based composites has become an urgent yet challenging task. X-based spectroscopic techniques are crucial tools for detecting changes in charge density upon interface formation [42]. For instance, X-ray absorption fine structure (XAFS) and X-ray photoelectron spectroscopy (XPS) can analyze the valence and local environment of interfacial interactions. Other techniques such as Kelvin probe force

microscopy and Raman spectroscopy can be employed to detect surface electronic properties and gain insights into the heterointerfaces of the composites [43].

3. Synthesis methods for CPs composites

3.1. Synthesis of CPs

In CPs, the carbon chains consist of alternating single and double bonds, known as alternating bonds, through which delocalized electrons generate electrical conductivity within the polymers [44]. The most common way to synthesize CPs is oxidative or electrochemical polymerization. Chemical polymerization possesses the advantages of creating various acceptor and donor units, ease of scale-up, suitable for p and n-type polymers, while showing less energy density and capacitance. Electrochemical polymerization is a single-step process for synthesizing the electrodes with a more porous surface area; however, some disadvantages such as unachievable huge oxidation potential to polymerize, uncontrollable molecular weight, limitations in reproducibility issues exist. In oxidative polymerization, CPs can achieve the desired molecular weight and long CP chains. Electrochemical polymerization achieves 1D CP films through deposition with high thermal stability at a single step. However, high concentrations of monomers were needed, and then due to strong electrolytes the film gets deteriorated during the electrochemical process [45].

In the process of synthesizing CPs, various oxidants including ammonium persulfate (APS), iron (III) chloride (FeCl_3), potassium permanganate (KMnO_4), potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$), as well as specific composites like Fe^{3+} /hydrogen peroxide (H_2O_2) and $\text{Cu}^{2+}/\text{O}_2$, are effective oxidants for the synthesis of PANI. In addition, using $\text{K}_2\text{Cr}_2\text{O}_7$ or FeCl_3 as oxidants for aniline polymerization resulted in a significantly low conversion rate. Fortunately, higher conversion rates can be achieved through combining with APS as a composite oxidant [46]. Transition metal compounds, such as Fe (III) and Cu (II) complexes, have been found to serve as polymerization initiators and dopants in converting pyrrole into electrical CPs. FeCl_3 is an oxidant that uses crucial reaction parameters such as reactant stoichiometry, temperature, and the reaction medium to form a stable PPy and PTh matrix [47,48].

Synthesizing PEDOT through oxidative chemical polymerization is comparable to the method used for preparing PPy. An important discovery was made in developing highly conductive PEDOT films using iron (III)-sulfonates as oxidants [49]. Since the sulfonates are soluble in typical organic solvents like ethanol and n-butanol, mixing ethylenedioxythiophene (EDOT) with Fe (III)-sulfonates could disperse PEDOT. In order to prepare highly conductive PEDOT, peroxides can also be utilized as substitute oxidants. In the second step to prepare PEDOT:PSS synthesis, the oxidative polymerization of PEDOT takes place in water with the help of a powerful oxidant (sodium persulfate or APS), in the presence of PSS matrix [50]. On the other hand, electrochemical polymerization of PEDOT:PSS can be synthesized in the absence of oxidants but with the electrolyte (lithium perchlorate (LiClO_4), 1-butyl-3-methylimidazolium hexafluorophosphate (BMIMPF_6), and lithium bis (trifluoromethanesulfonyl) amide (LiTFSI)). During electropolymerization, the anions from the electrolyte are incorporated into PEDOT as counterions to maintain a stable charge [51]. Finally, PIn is commonly synthesized using APS and FeCl_3 as an oxidant in oxidative polymerization, whereas, in electrochemical polymerization, $\text{CH}_3\text{CN}/\text{LiClO}_4$ and water/ LiClO_4 electrolytes were used [52].

The standard synthesis schemes for PANI [53], PPy [54], PTh [55,56] and PIn [57] from their respective monomers and oxidative reagents are illustrated in Fig. 2. The polymerization process typically comprises three stages: oxidation, coupling, and propagation [58]. Initially, the monomer undergoes oxidation to form a radical cation, which then combines with another radical cation to form a dimeric radical cation. Subsequently, this dimeric cation attracts another monomer, leading to the formation of a trimeric radical, and this process

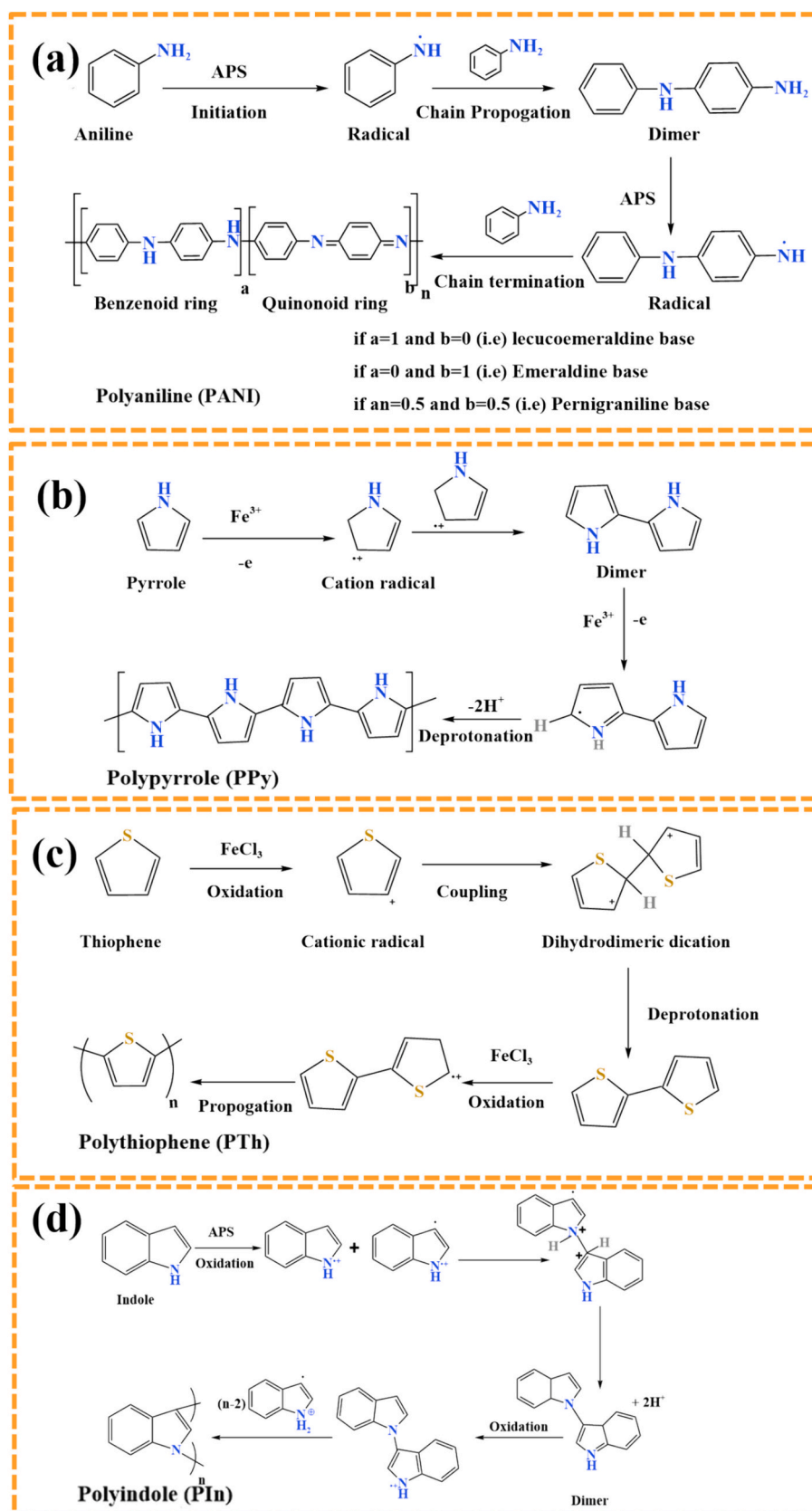


Fig. 2. Polymerization of a) PANI. [53] b) PPy. [54] c) PTh. [55,56] and d) PIn [57] from the monomer and oxidative reagents.

continues iteratively to elongate the polymeric chain. The doping state, size, dimensions, and structural confinement of CPs are influenced by factors such as solvent choice, surfactant selection, and temperature control [59,60].

3.2. Synthesis of CPs composites

CP composites can be synthesized through diverse methods, with the most commonly employed techniques including solution polymerization-based methods, deposition-based methods, and hydrothermal polymerization [61,62]. Factors such as reaction temperature and time, surfactants or dopants, and integrating additives can regulate the composite's morphology, the interface affinity and interface bonding strength between CPs and additives [63]. Mechanical mixing parameters like speed, duration, and shear force impact CP dispersion and alignment. Electrochemical deposition parameters such as voltage, current density, and electrolyte composition dictate morphology of the CP film [45]. This section delves into widely utilized methods for preparing CP composites for SCs.

3.2.1. Solution polymerization-based methods

Solution polymerization-based methods possess the advantages of homogeneous reaction environment, contrabability, ease of scale-up, involving two primary steps: (i) dispersing the components (CPs and additives) in a suitable solvent (organic or aqueous medium) through ultrasonication or magnetic stirring, and (ii) evaporating or removing the solvent to yield composite powder, thin films, or fibers [59]. Techniques such as vacuum filtering, chemical oxidation, solution casting, printing, and fiber-spinning are all encompassed within solution processing methods utilized for fabricating film-type CP electrodes.

The underlying principle of this approach typically involves dissolving the organic monomer and additives network in an acidic environment (solution A) and dissolving the oxidizing agents under similar conditions (solution B) to generate precursors for the polymer-carbonaceous composite. The polymerization process is initiated by gradually adding solution B to solution A, followed by allowing the mixture to stand for several hours. Subsequently, the composite undergoes drying in a hot air oven after washing to eliminate any residual monomer and additives.

3.2.1.1. Fiber spinning. Fiber spinning is a versatile and efficient method for producing synthetic fibers with tailored properties continuously. In this method, an additive and an extruded liquid CPs are utilized to continuously draw and solidify the mixture, forming composite fibers (threads). The core principle of solution spinning relies on the evaporation of solvent while the solution stream undergoes ultrafine stretching by a swiftly moving gas flow, resulting in the formation of nanofibers. For instance, polyacrylonitrile (PAN) nanofiber core-spun yarns are created by enveloping PAN nanofibers with cotton yarns via solution jet spinning (Fig. 3a). Furthermore, in-situ polymerization is employed to deposit PPy particles onto the nanofiber core yarns, thereby fabricating layered electrodes comprising PPy/Polyacrylonitrile (PAN)/cotton yarns [64]. Additionally, in a research by Yang et al., [65] CNT@PANI yarn is produced through fiber spinning subsequent to the preparation of the composite CNT@PANI via the in-situ polymerization technique. The process of preparing rGO (reduced graphene oxide) + PANI@CMC@rGO + PANI (CMC-carboxymethyl cellulose) fibers involves pouring the GO/vitamin C/PANI mixture into a syringe and injecting it into a rotatable coagulation bath to form the composite fiber [66]. In another study focusing on PEDOT:PSS-based composite fibers, carbon black and CNTs are initially wet-spun, followed by dip-coating with PEDOT:PSS to enhance capacitance and electrical conductivity [67]. This fiber-spun CP composite method exhibits a greater specific surface area and more uniform pore size distribution compared to conventional cotton yarn. This characteristic provides an increased number

of active sites for active ingredients. The composites prepared from the fiber spinning have more channels for electrolyte penetration, high resistance, and good bending with high specific capacitances.

3.2.1.2. Vacuum filtration. Vacuum filtration is a common laboratory technique used to separate solids from liquids using a vacuum to draw the liquid through a filter paper or membrane, which is time-saving, compatible with different solvent and controlled. This method distinguishes itself as one of the simplest approaches for preparing thin film CPs composites, ensuring a uniform and compact distribution. For example, Liu et al. demonstrated the preparation of homogeneous PEDOT:PSS/aramidnanofibers (ANFs) by mechanically agitating an ANFs solution with PEDOT:PSS. The resulting PEDOT:PSS/ANFs solution was then layered onto a 0.22 μm -pore-sized nylon microporous filter membrane and vacuum-filtered for 7–8 h to form a wet PEDOT:PSS/ANFs film (Fig. 3b). Subsequently, the wet composite film was removed from the nylon microporous filter membrane by immersion in methanol and rinsed three times with isopropanol solvent [68]. Liang and colleagues prepared the PANI:PSS/Carbon nanofiber (CNF) composite via in-situ filtration using aniline, PSS, and carbon nanofibers (CNF). Furthermore, the composite film was prepared by vacuum-filtering this solution using a polypropylene filter. As per the author, polypropylene with a three-dimensional (3D) hierarchical porous structure through vacuum filtering increased the mechanical strength, flexibility, and electrolyte absorption of PANI:PSS/CNF, concurrently reducing production costs. [69]. Similarly, PPy nanofibers and MXene were physically shaken, and the solution was filtered through vacuum filtration to form PPy/MXene nanofilms [70]. This method was chosen to reduce MXene self-stacking by intercalating PPy nanofibers and mitigating PPy swelling.

3.2.1.3. Printing method. This technique offers a cost-effective method for preparing electrodes, where additive components are dissolved in a CPs solvent to create a functional ink, subsequently printed onto the substrate. Through this process, the functional ink permeates the substrate via the mesh of a stencil. Illustrated in Fig. 3c is the preparation of composite using PEDOT/manganese oxide (MnO_2) by Li et al. [71]. Initially, they printed PEDOT:PSS onto polyester fabrics and allowed it to dry for 15 mins at 120 $^{\circ}\text{C}$. Then, electrodeposition was conducted using sodium sulfate and manganese sulfate as electrolytes with the printed PEDOT:PSS serving as a working electrode. The resulting MnO_2 , generated from the electrochemical reaction, was electrodeposited onto the PEDOT:PSS to form the composite, utilized as the highly flexible CPs composite electrode. The unique construction enables the utilization of thick electrodes containing an abundance of active sites and interpenetrating fiber networks with high conductivity. Similarly, a PANI-based composite electrode was fabricated using 3D printing, where GO was initially self-assembled in a mixture solvent of N-methyl-2-pyrrolidinone and water. These inks offer shaping ability, self-sustainability, and electrical conductivity following GO reduction, forming the flexible PANI/GO composite film. The synergy between PANI and GO presents a distinctive opportunity to formulate customizable and printable ink for the 3D printing of PANI. [72]. For the fabrication of a stretchable rGO-CNT-PEDOT:PSS flexible composite, Zhou et al. utilized aerosol jet printing with a methanol solution. The addition of CNTs and PEDOT:PSS enhances performance by reducing rGO restacking and promoting good conductivity [73].

3.2.2. Deposition-based methods

Composite structures comprising CPs matrix/additive networks or metal composites were also achieved via deposition onto suitable substrates [74,75]. This process involved depositing the substance either directly onto the substrate or onto a previously deposited layer. A critical aspect of these techniques was the precise control of layer or film thickness. Commonly employed deposition procedures in constructing

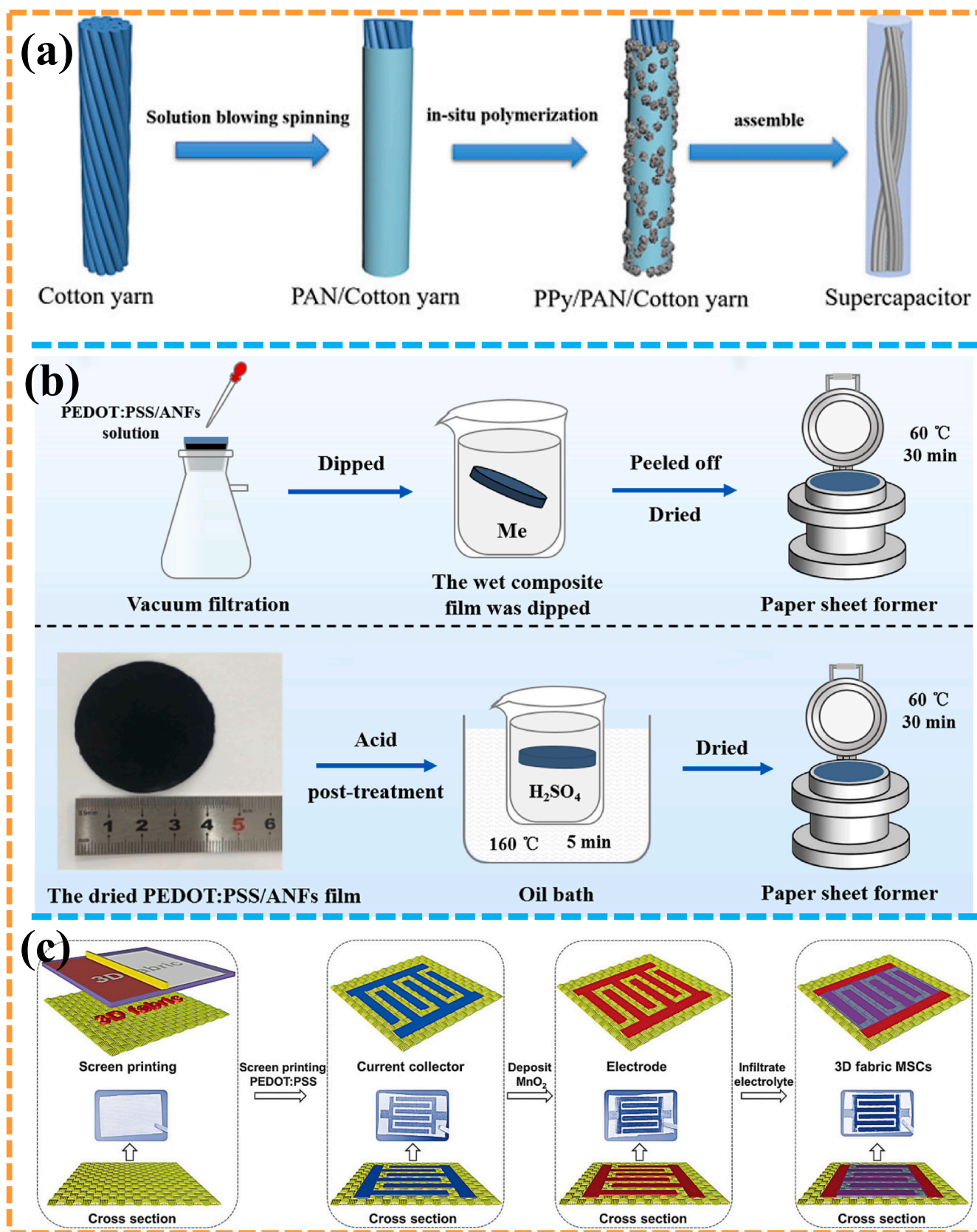


Fig. 3. Graphical illustrations of strategies to prepare CP composites using solution types of (a) Fiber spinning [64]. (b) Vacuum filtration. [68] (c) Printing method. [71].

composites composed of CPs and additive networks are included. These techniques offer additional advantages, including facile control over pore volume and size, along with a reduced number of synthesis steps compared to alternative methods.

3.2.2.1. Electrochemical deposition. Electrochemical deposition, also known as electrodeposition or electroplating, is a process that involves the deposition of a coating with controlled thickness, composition and properties onto a substrate by using an electric current. This method stands as a widely employed, simple, and efficient approach for preparing CPs composite electrode materials, serving as a cornerstone in synthesizing various composite nanostructures [76]. For instance, Li et al. synthesized a CPs-based ternary composite CNT film-iron oxide (Fe_3O_4)-PANI using the electrodeposition method (Fig. 4a). The working, counter, and reference electrodes comprised the CNT film- Fe_3O_4 composites, a $20 \times 20 \text{ mm}^2$ platinum sheet, and an Ag/AgCl electrode, respectively. Cathodic electrodeposition was carried out at a constant potential of 0.75 V in a mixed solution containing 0.45 M H_2SO_4 and 0.25 M aniline, leading to the intimate bridging of PANI between adjacent particles. This method influences the morphology of the composites and maintains the geometric pattern and crystallinity of the additives [77]. Similarly, a PEDOT/graphite composite electrode was prepared via the electropolymerized potentiostatic technique (1 V) on graphite substrate, utilizing 3,4-ethylenedioxythiophene (EDOT) as the monomer, FeCl_3 as the supporting electrolyte, and chlorine (Cl) as the doping agent. This deposition technique enhances electrical contact between the PEDOT polymer and the current collector, leading to low contact resistance. This facilitates smooth charge movement at the interface between the polymer and substrate. [78]. Furthermore, Raj et al. fabricated a $\text{PIn}/\text{Co}_3\text{O}_4$ composite through galvanostatic electrochemical polymerization (450 and 700 mA cm^{-2} for three mins) in an electrolyte consisting of 0.01 M HNO_3 , 0.05 M $\text{Co}(\text{NO}_3)_2$, and 0.25 M NaNO_3 with the indole monomer [40].

3.2.2.2. Electrospray deposition. The electrospray deposition method is a versatile technique used to deposit thin films or coatings onto substrates with precise control over thickness and composition. Typically, the monomer and additive compounds are initially dispersed in an acidic medium before being sprayed onto the substrate through a nozzle under pressure. Upon deposition, the spray solution undergoes polymerization, resulting in the formation of the composite film. For instance, Dhandapani and their team synthesized a thin layer of PPy/carbon quantum dots (CQDs) via electrospray deposition on Ni foam substrate. Uniform thin PPy films were achieved by applying a voltage of 8 kV and pneumatic pressure ranging from 0 to 3 bar, with a 15 mm distance between the substrate and the nozzle tip (Fig. 4b). The resulting composite, prepared using the electrospray method, exhibited a uniform distribution of particles on the substrate with consistent thickness [79]. Similarly, the same group fabricated a $\text{PIn}/\text{carbon dots (CDs)}$ composite film by electrospraying a solution containing PIn and nitric acid onto a conductive substrate along with CDs [80]. Moniz and colleagues prepared an electrodeposited PEDOT:PSS/carbon composite, adjusting parameters such as distance (4–10 cm), applied voltage (12–18 kV), flow rate (40–80 $\mu\text{L h}^{-1}$), and deposition periods (5, 10, 20, 30, and 60 min). These parameters produced different morphologies with micro-size droplets made effective for coating PEDOT:PSS on carbon yarn that form a porous and created high charge storage [81]. Furthermore, Iqbal et al. developed a membrane using carbon black-doped carbon nanofibers (CB@CNF) by carbonizing polybenzoxazine as a precursor at 800 °C. Subsequently, PANI was electrodeposited onto the membrane surface, enhancing its flexibility, inhibiting fracture, and forming a core-shell structure [82].

3.2.2.3. Chemical vapor deposition (CVD). CVD is a process used to produce high-quality, high-performance solid materials, typically thin

films, possessing characteristics of controllability, versatility and efficiency. This technique involves exposing volatile precursors to the substrate to produce CPs composite films. Depending on the application, this method proves to be more practical and preferred for fabricating composites in various forms such as powder, film, or fiber. Recently, Smolin et al. employed an oxidative-type CVD technique to synthesize PANI/CDC composites, where aniline polymerization with carbide-derived carbon (CDC) occurs in a solvent-free single step, resulting in an evenly coated composite with a 1–2 nm increase in thickness. Fig. 4c illustrates the preparation of PANI/CDC via the CVD method. The author used this method due to the involvement of polymerization in the surface and facilitating uniform coating and controlling the thickness [83]. In a separate study on PPy/ MnO_2 @CF, a vacuum chamber containing MnO_2 @CF with Fe^{3+} ions deposited on its surface was utilized. After introducing the pyrrole monomer into the vacuum chamber and reaching a pressure of 0.1 atm at 80 °C, the monomer was transformed into gas, and the reaction continued for 20 mins to form the PPy/ MnO_2 @CF composite. Additionally, Lee and co-researchers prepared graphene/dopamine/PPy nanocomposites on a polymer substrate with the aid of dopamine. Initially, graphene was coated with dopamine and FeCl_3 and then inserted into a glass device for CVD polymerization with the pyrrole monomer. After 30 mins of deposition, a layer of PPy formed on the film, rendering the composite suitable for flexible SCs [84].

3.2.3. Hydrothermal technique

Hydrothermal method is a synthesis technique using high-temperature, high-pressure and uniform heating to facilitate chemical reactions and formation of material with unique structure controllably. In the hydrothermal technique, CPs composites can be prepared through two methods: either polymerizing the monomer first to form CPs and then reacting with additives in the hydrothermal process, or polymerizing directly within the autoclave. Among these, CPs are typically prepared through oxidative polymerization and then combined with additives in the hydrothermal reaction. For instance, Zhang et al. employed the hydrothermal technique to fabricate a PPy/graphene electrode on a carbon cloth substrate in a single step. The solution mixture containing pyrrole monomer and graphene was applied to the carbon cloth, undergoing a redox reaction in the hydrothermal process (Fig. 4d). This technique enhanced the specific surface areas and produced remarkable electrochemical performances [85]. Similarly, PANI/graphene composites were synthesized by mixing GO solution, aniline, and concentrated HCl with APS and stirring for 4–6 h. The resulting solution was then transferred to autoclaves for a 6-h hydrothermal reaction at various temperatures [86]. Building on this approach, Lei and colleagues synthesized graphene/PEDOT by homogeneously mixing EDOT, GO suspension, and FeCl_3 in an ethanol- H_2O co-solvent (1:1 by volume). After 10 mins of bath sonication, the resulting slurry was poured into a Teflon-lined autoclave and heated for 12 h at 180 °C. The optimization studies revealed that increasing the temperature will lead to the aggregation of additives in polymers. Meanwhile, the hydrothermally synthesized composite exhibited high reversibility with good capacitance [87]. Alternatively, Chao et al. synthesized PEDOT:PSS/ MoS_2 composites via the hydrothermal technique using L-cysteine and $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$ (sodium molybdate dihydrate) in a PEDOT:PSS dispersion, heated to 180 °C for 24 h. A cylindrical hydrogel MoS_2 /PEDOT:PSS composite was synthesized and collected once cooled to room temperature. The ability of CPs to be processed in solutions through complexation/chelation and electrostatic interactions with additives and avoids aggregation. This motivates the authors to investigate polymers that act as stabilizers and assist in the hydrothermal synthesis of MoS_2 that are easily dispersible [39].

4. Effects of interface engineering in CPs composites

When composite materials are framed, the interfacial interaction mainly exist in the form of noncovalent interactions (i.e, hydrogen

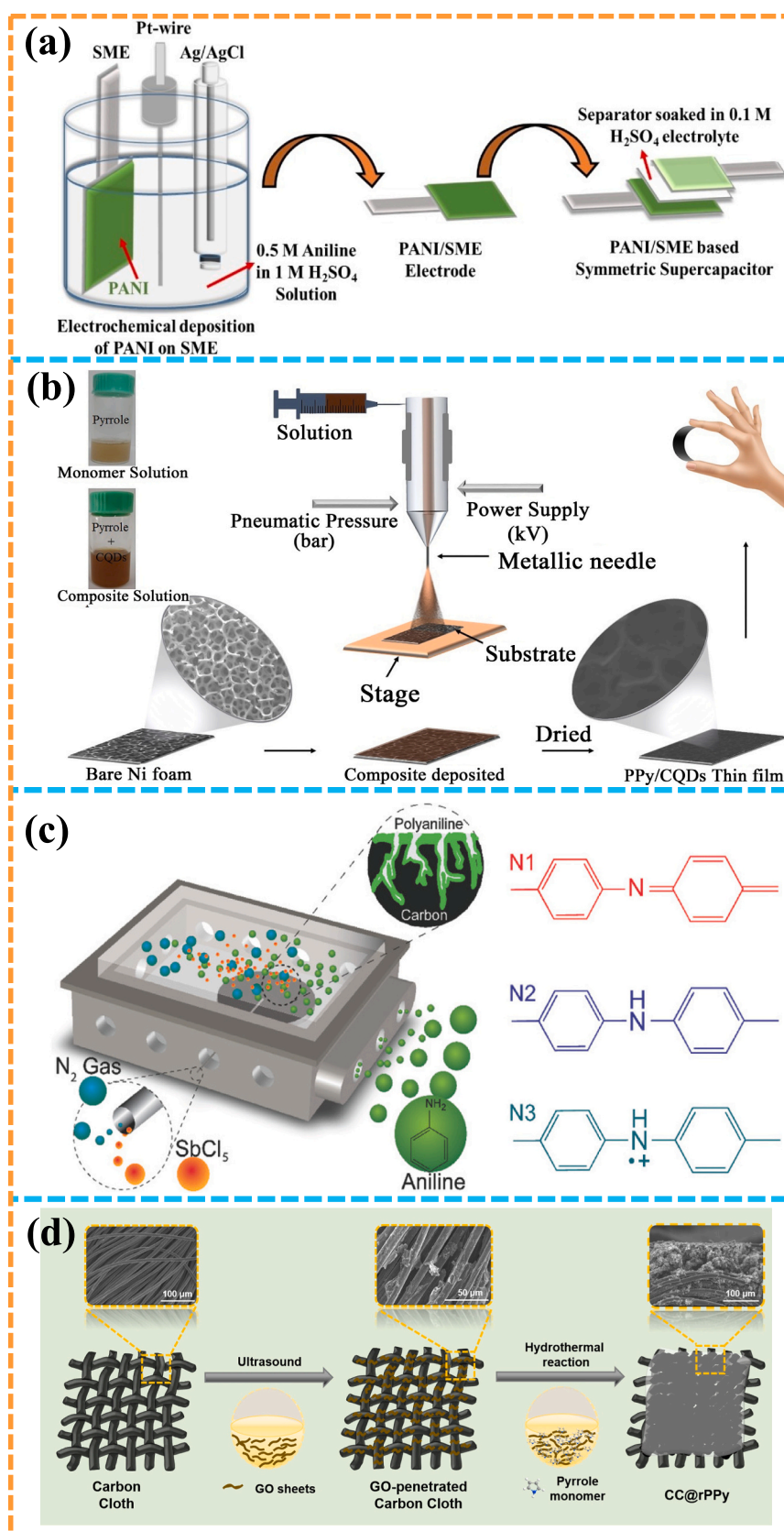


Fig. 4. CP composite preparation with (a) Electrochemical polymerization. [77] (b) Electrospray deposition. [79] (c) Oxidative chemical vapor deposition. [83] (d) Hydrothermal reaction. [85].

bonding [88], electrostatic, van der Waals), affecting the charge transfer greatly [89]. More importantly, the synergetic effect owing to the coupling of the advantageous characteristics of a single material either CPs or carbon/metal oxides yields a great positive influence on the SCs performances [90]. In the following sections, we discussed about the positive impact of non-covalent interactions, confinement effect and synergetic effect on the electrochemical performances (specific capacitance, rate capability and stability).

4.1. Non-covalent interactions

Non-covalent interactions exist widely in CPs-based systems, connecting more individual components. Strongly directed noncovalent interactions mainly include hydrogen bonding, electrostatic force, π - π stacking, and other polar interactions. These interactions are reversible as opposed to covalent bonding, allowing for flexible construction and reconstruction of the composites, and exhibits a positive influence on the electrochemical performance.

4.1.1. Hydrogen bonding

Intimate interfaces can be established through hydrogen bonding, which exhibits considerable binding strength ranging from 12 to 120 kJ/mol [91]. The specificity and selectivity of hydrogen bonds contribute to molecular recognition between CPs and additives. Hydrogen bonds form between polar groups containing an atom with a free pair of electrons and a covalently bound hydrogen atom. Typically, hydrogen bonds occur among nitrogen and oxygen groups, representing strong bonds between electron-rich atoms (such as N and O atoms in polar groups like -COOH, -OH, -NH₂, and -NH₃, etc.) and electron-deficient hydrogen atoms within these groups, leading to the formation of hydrogen bonds [92].

Due to the omnipresence of hydrogen bonds in CPs-based composites, understanding the effects of hydrogen bonds is important before delving into other non-covalent interactions, such as electrostatic and π - π bonding. Understanding the role of hydrogen bonds takes

precedence as they predominantly influence the elongation of CPs nanostructures. This is because a hydrogen atom, when coupled with a free electron pair of a highly electronegative atom (such as nitrogen, oxygen, or fluorine), results in a robust dipole-dipole interaction. [93]. In carbon materials, hydrogen bonding always coexists with other non-covalent interactions, playing a vital role in composites like g-C₃N₄ and CPs composites. This addition enhances electrolyte wettability, surface polarity, electrical conductivity, and electron-donor characteristics, resulting in effective charge transfer and exceptional long-term cycle stability [94,95]. Unfortunately, the Van der Waals force leads to the easy restacking of single-layered g-C₃N₄, thereby reducing specific power and energy density. Consequently, combining g-C₃N₄ with CPs might offer a strategic solution to overcome these limitations. For example, bands (3212 cm⁻¹) observed in the IR spectrum and shifts in peaks indicate the presence of hydrogen bond interactions between g-C₃N₄ and PANI (Fig. 5a). Moreover, as evidenced by the electrochemical results, compared to pure PANI, the PANI/g-C₃N₄ composite prepared via in-situ oxidative polymerization exhibited improved electrochemical performance, showing a specific capacitance of 584.3 F/g and 81.91 % capacitance retention at 1 A g⁻¹ after 1000 cycles (Fig. 5b, c). The findings demonstrate the establishment of PANI hydrogen bonding interactions with g-C₃N₄, which could positively impact the stability of the composite [96]. Furthermore, higher specific capacitance values were observed (475 F/g at a current density of 0.5 A g⁻¹) due to interface hydrogen bonding, in contrast to pure g-C₃N₄ and PPy, where both blocks exhibited capacitance values below 100 F/g. The porous structure of the composite facilitates ample contact between the electrolyte and active material, thereby expediting the charge-transfer reaction. This composite also demonstrated stability of 84 % of retention after 500 cycles [97]. The incorporation of Ag introduces exceptional properties, including outstanding electrical and thermal conductivities, as well as high thermal stability. Thus, the Ag/PANI/g-C₃N₄ composite was formulated to enhance cycling durability and electronic conductivity while minimizing g-C₃N₄ accumulation. To further bolster the stability of PANI/g-C₃N₄, researchers developed modified composites

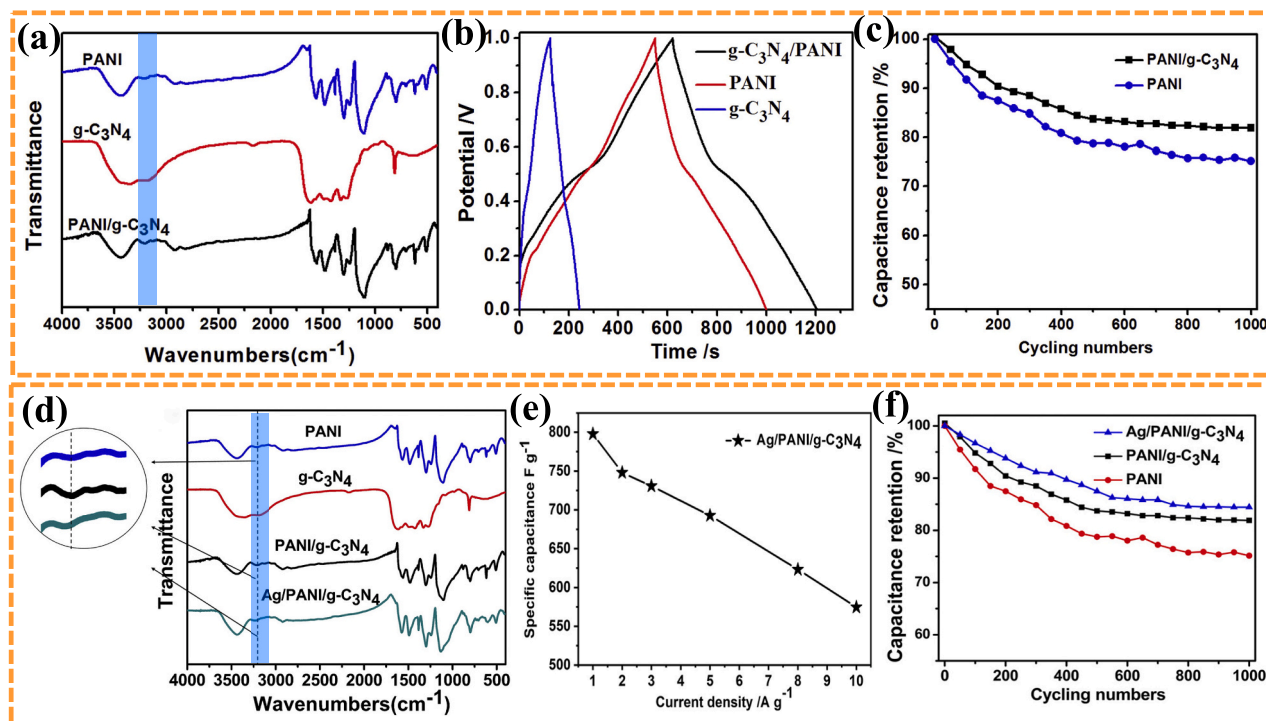


Fig. 5. (a) FT-IR spectra of g-C₃N₄, PANI and PANI/g-C₃N₄. (b) Galvanostatic charge/discharge curves of g-C₃N₄, PANI and PANI/g-C₃N₄ at 1 A g⁻¹ in 1 M H₂SO₄ solution. (c) Cycling performance of PANI and PANI/g-C₃N₄ at 1 A g⁻¹. [96] (d) FT-IR spectra of PANI, PANI/g-C₃N₄, g-C₃N₄, Ag/PANI/g-C₃N₄. (e) Specific capacitance graph of Ag/PANI/g-C₃N₄ at various current densities. (f) Cycle stability up to 1000 cycles for PANI, Ag/PANI/g-C₃N₄, PANI/g-C₃N₄ at 1 A g⁻¹. [37].

incorporating with Ag metal nanoparticles. Experimental findings revealed a strengthening of hydrogen bonding intensity within PANI and the formation of interactions between PANI and g-C₃N₄, as confirmed by FTIR peaks in the composites (Fig. 5d). This interaction holds the potential to enhance composite stability, a notion supported by electrochemical results. Remarkably, after 1000 cycles at 1 A g⁻¹, the Ag/PANI/g-C₃N₄ composite displayed a high specific capacitance of 797.8 F/g and exceptional capacitance retention of 84.43 % and (Fig. 5e, f).

This underscores the positive impact of hydrogen bonding, facilitated by the combined EDLC nature of g-C₃N₄ and pseudocapacitive PANI [37].

Additionally, the presence of hydrogen bonds can mitigate the formation of a dense PPY structure, potentially leading to the generation of more electrochemically active sites and significantly increasing the utilization efficiency of PPY. For instance, pyrrole underwent chemical polymerization on polyacrylic acid (PAA) functionalized carbon fabric (FCF) to fabricate FCF/PPy composite. Chemical interactions in these

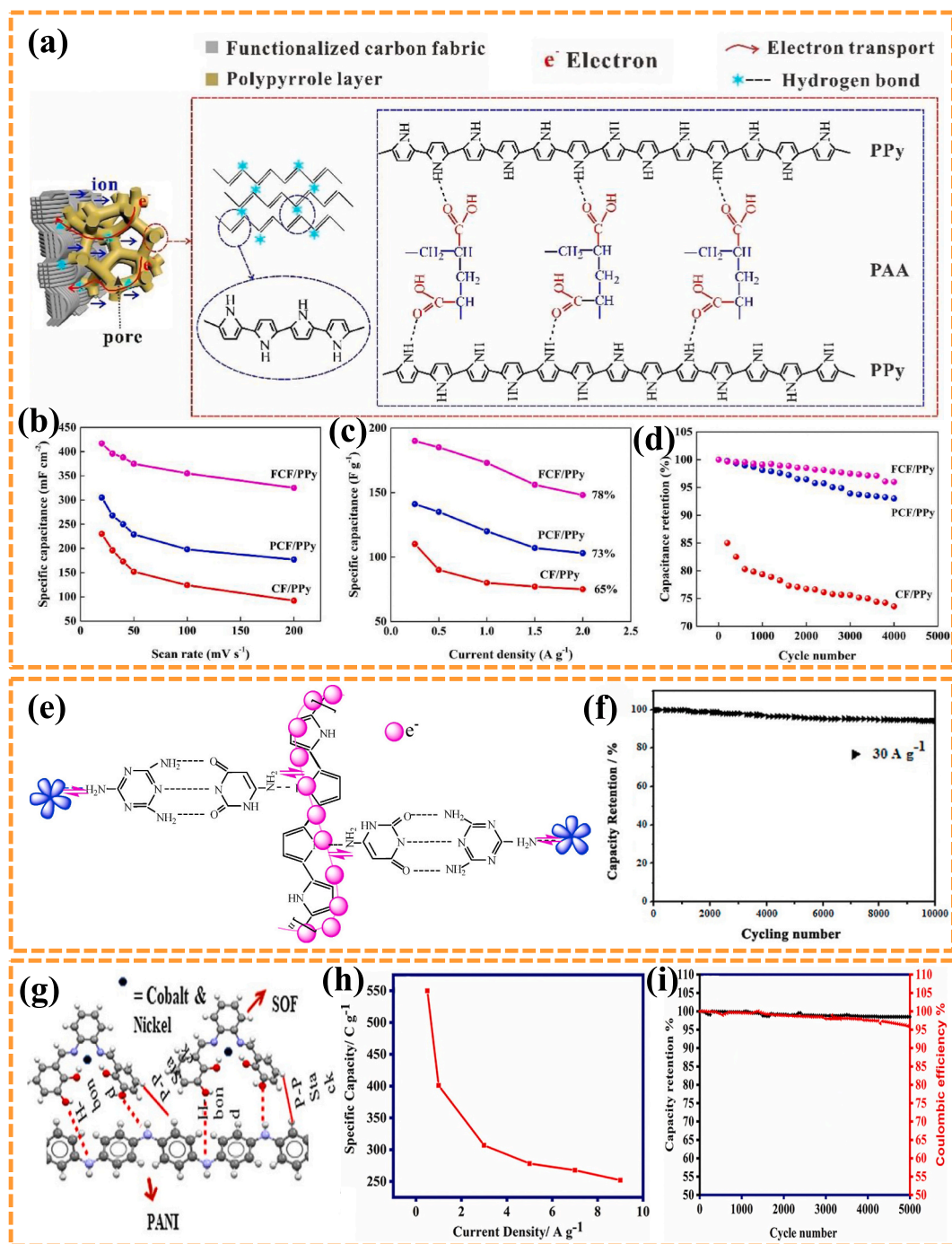


Fig. 6. (a) Schematic representation of highly efficiency utilization of PPy based on the design idea of the hydrogen bond strategy polymerization. (b) Specific capacitances of FCF/PPy, PCF/PPy and CF/PPy at a scan rate range of 20 mV s⁻¹ - 200 mV s⁻¹. (c) Specific capacitances of FCF/PPy, PCF/PPy and CF/PPy at a current density range of 0.25 A g⁻¹ - 2.0 A g⁻¹. (d) Capacitance retention of FCF/PPy, PCF/PPy and CF/PPy in 4000 charge/discharge cycles at 2 A g⁻¹. [98] (e) Illustration of electron transportation in the fabricated electrode. (f) Cycling performance of PPy@NiCo(OH)₂ at 30 A g⁻¹. [101] (g) the schematic diagram of interaction in Ni-Co@PS. (h) Specific capacitance of Ni-Co(3:1)@PS-CC electrode at different current densities. (i) Cycling performance of Ni-Co(3:1)@PS-CC electrode at 9.0 A g⁻¹. [102].

PPy composite electrodes may involve hydrogen and ionic bonds. The schematic representation of the highly efficient utilization of PPy in the hydrogen bond strategy is given in Fig. 6a. As indicated by the N1s spectra from XPS analysis, the effective hydrogen bond between PPy-COOH and -NH resulted in the increase in Δf for FCF/PPy. The remarkably efficient utilization of PPy yielded a high areal specific capacitance of approximately 417 mF cm^{-2} from cyclic voltammetry (CV) (Fig. 6b) and a specific capacitance of 190 F/g (Fig. 6c). On the other hand, the hydrogen bond constrained the mobility of PPy chains on the substrate, leading to an impressive 96 % capacitance retention for FCF/PPy after 4000 charge/discharge cycles (Fig. 6d). Furthermore, the aqueous devices exhibited a high energy density of 16.9 Wh kg^{-1} at a power density of 100 W kg^{-1} , attributed to the effective ion and electron transfer facilitated by the unique structure of the PPy ordered chain, loose crosslinked nanostructure, and interfacial hydrogen bonds [98]. The improved electrochemical stability observed in PPy electrodes implies that the interfacial hydrogen bonds effectively reinforce the PPy chain, thus attenuating damage during cycle stability tests.

In addition to the aforementioned carbon materials, researchers are deeply fascinated by the extraordinary spatial structure of nickel-cobalt dihydroxide ($\text{NiCo}(\text{OH})_2$), endowing it with remarkable chemical and physical properties. These properties encompass ion exchange, thermal stability, and conductivity. Moreover, the structure facilitates a larger specific surface area, which proves for the EDLC formation. Concurrently, the variable valence anions in the laminate contribute to an augmented Faraday quasi-capacitance and a significant proliferation of electrochemically active sites. Nonetheless, $\text{NiCo}(\text{OH})_2$ often faces challenges such as reduced capacitance and unstable cycling, attributed to structural destruction incurred during high-current charge and discharge cycles [99,100]. For example, the combination of $\text{NiCo}(\text{OH})_2$ and PPy is facilitated through hydrogen bonding between nitrogen and hydrogen, utilizing a three-hydrogen bond system formed by thymine and melamine, as depicted in Fig. 6e. The deposition of PPy effectively enhances the conductivity of $\text{NiCo}(\text{OH})_2$ while promoting Faraday processes on both sides of the interface. The unique π - π conjugate structure within the three-hydrogen bond system serves as a pathway for rapid electronic and ionic transport, as illustrated in Fig. 6f. Consequently, leveraging the hydrogen bond, the resulting electrode material achieves a capacitance of 1469.25 F/g at 1 A g^{-1} , showcasing commendable electrochemical performance. [101]. A pair of redox peaks from -0.2 – 0.6 V indicate Faraday redox processes of $\text{PPy@NiCo}(\text{OH})_2$, designated for the Faradic reaction between $\text{Ni}^{2+}/\text{Ni}^{3+}$ and $\text{Co}^{2+}/\text{Co}^{3+}$ with OH^- . The capacitance retention exhibited 95 % after 10,000 cycles at the current density of 30 A g^{-1} .

Apart from conventional CPs, polymeric transition metal complexes such as salen and salphen (SOF) also exhibit pseudocapacitive properties suitable for energy storage. Schiff base ligands derived from salicylaldehyde, and its derivatives are employed in the synthesis of these materials. Moreover, numerous researchers have detailed the synthesis of SOF with π -conjugated ligands situated between coordination sites and metals [102]. For example, the Adriyani group synthesized a composite of Ni–Co nanoparticles and PANI-salphen (Ni-Co@PS) using an in-situ polymerization technique. The schematic interaction within Ni-Co@PS is illustrated in Fig. 6g. This composite structure, featuring hydrogen bonds and stacking bonds between PANI and the metal salphen complex, demonstrated a high specific capacitance of 549.994 C g^{-1} (1447.2 F/g , 0.5 A g^{-1}), as depicted in Fig. 6h. Additionally, the Ni-Co(3:1)@PS electrode exhibited prolonged cyclic stability, retaining 95.9 % of its capacitance after 5000 cycles, as shown in Fig. 6i [102].

Dhandapani et al. synthesized the PIn/g- C_3N_4 composite electrode using the reflux method [103]. Their findings reveal that the composite boasts high energy and power densities, balanced porosity, and improved ionic transport facilitated by hydrogen bonding between g- C_3N_4 and the NH groups of PIn. The higher pore volume of $0.089 \text{ cm}^3/\text{g}$ for PIn/g- C_3N_4 enhances accessibility for the diffusion of electrolyte ions into the interior surface, thereby accelerating oxidoreduction reactions.

PIn/g- C_3N_4 exhibit a specific capacity of 440.8 C/g at a current density of 6 A g^{-1} , retaining 72 % of this capacitance after 5000 cycles. These promising results suggest that PIn/g- C_3N_4 composites hold great potential as electrode materials for future commercial energy storage applications. Furthermore, PIn demonstrates strong hydrogen bonding capabilities with MXene structures. De and colleagues confirmed this hydrogen bonding via photoluminescence (PL) analysis, highlighting its role in facilitating electron transfer between conductive PIn and MXene, thereby enhancing the electrochemical activity of SCs. Morphological analysis confirms successful intercalation and coating of PIn on MXene nanosheets. Optimizing the PIn concentration is crucial for achieving the best electrochemical performance in conjunction with MXene. Surface area analysis indicates that the MXene/PIn composite system with a 3:1 ratio of MXene to PIn exhibits the highest specific surface area, leading to a specific capacitance of 226.5 F/g at a current density of 2 A g^{-1} . Moreover, this nanocomposite demonstrates outstanding cyclic stability, retaining 90.5 % of its capacitance after 8000 cycles [34].

From the preceding discussion, it's evident that hydrogen bonding plays a pivotal role in CPs-based systems incorporating functionalized carbon materials (such as g- C_3N_4 and carbon fiber) or layered metal hydroxides, uniquely enhancing SCs performance. PANI, PPy and PIn are notably influenced by hydrogen bonding with additives during the synthesis, preventing the formation of dense polymer structures and avoids the agglomeration of additives. This interaction between CPs and additives through hydrogen bonding enhances the resulting material's thermal stability, conductivity and cycle stability which ultimately leading to superior electrochemical performance in SCs.

4.1.2. Electrostatic interaction

Currently, numerous researchers have reported the attachment of anionic CPs to positively charged nanocarbons through electrostatic interactions [104–106]. However, cationic functionalization of nanocarbons requires initial processing, typically achieved through two methods: (a) covalently attaching cations to graphene or oxidized CNTs, utilizing available -OH and -COOH groups for further functionalization; and (b) employing intermolecular interactions [106]. Subsequently, electrostatic interactions within CPs composites can occur between the cationic groups in carbon and the anionic groups in CPs [3].

As an illustration, composites comprising CPs and QDs have been formulated as active electrode materials to bolster electrochemical stability in SCs. QDs, graphite particles smaller than 10 nm , exhibit distinctive traits including outstanding solubility in water, robust electrical conductivity, and swift electron transfer properties. Leveraging an electrostatic self-assembly technique, Jian and collaborators crafted CQD-reinforced PPy nanowires (PPy-NW). [107]. The fabricated device (Fig. 7a), incorporating the electrostatic interaction between PPy's positive charge and CQDs' negative charge, along with high conductivity, demonstrates improved electrochemical reactivity. The intricate mechanisms underlying this phenomenon attribute the facile dispersion of positively charged PPy-NW (with a zeta potential of $+51.7 \text{ mV}$) in water. The electrostatic interaction between CQD and PPy were confirmed by infrared region at 1714 cm^{-1} . The composite electrode demonstrates pseudocapacitive behavior, achieving a specific capacitance of 306 F/g at a current density of 0.5 A g^{-1} (Fig. 7b). Furthermore, the CQDs/PPy-NW composite maintains a capacitance retention of 85.2 % after 5000 cycles, indicating the enhancement of electrostatic forces on the stability of CPs composite. Due to its substantial specific area, augmented active sites, and exceptional electronic conductivity, the as-prepared CQDs/PPy-NW composite exhibits remarkable electrochemical characteristics.

To advance SCs electrode development for high-energy storage applications, ternary composites (SGP3) comprising stannic oxide QDs (SnO_2 QDs), GO, and PPy were investigated [108]. These materials possess unique characteristics such as cyclic stability and a large specific surface area. SnO_2 nanocrystals, exhibit exceptional thermal stability, electrical conductivity, and high capacitance. However, the

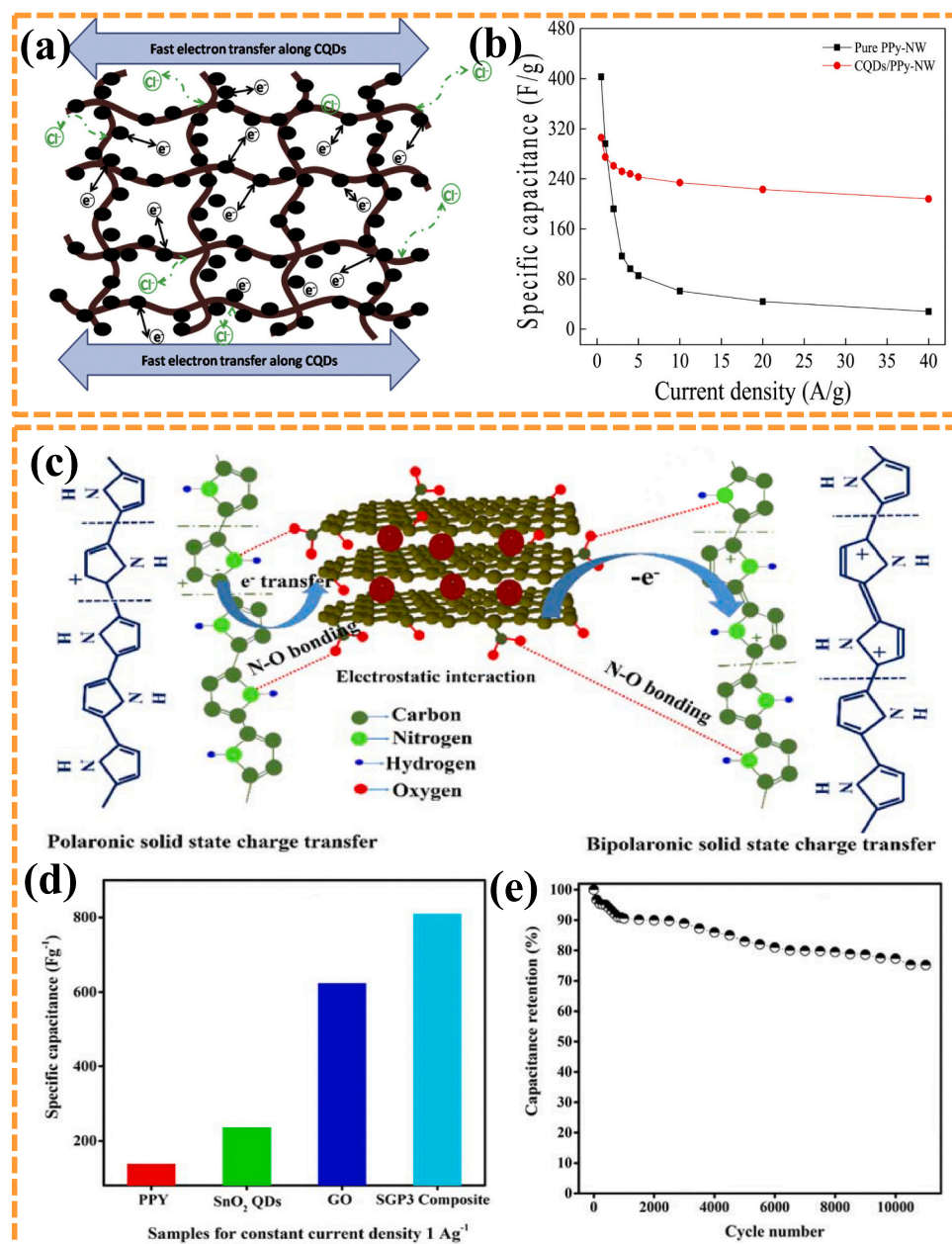


Fig. 7. (a) Schematic diagram demonstrating the capacitive properties for CQDs/PPy-NW composite. (b) Specific capacitance of PPy-NW and CQDs/PPy-NW composite electrodes at different current densities. [107] (c) Systematic diagram of the electron transfer in SGP ternary composite. (d) Specific capacitance of pure PPy, SnO₂ QDs, GO and SGP3 composite at 1 A g⁻¹. (e) Cycle stability of SGP3 device up to 11,000 cycles at 7.4 A g⁻¹. [108].

performance of such materials is often impeded by resistance and instability. When combined with PPy, the typical pseudocapacitive material SnO₂ enhances charge storage capabilities. Fig. 7c illustrates electron transfer mechanism facilitated by electrostatic interaction in SGP ternary composite. From the transition electron microscopy (TEM) author conveys the observed distinctive array of black dots representing SnO₂ quantum dots dispersed across the graphene sheet, attributed to the strong electrostatic contact between SnO₂ QDs and GO. This is supported by the O1s XPS spectrum, which reveals the presence of C=O and C-OH bonds, facilitating the dispersion of SnO₂ QDs on the graphene nanosheets through electrostatic interaction. Due to enhanced physicochemical properties, the SGP ternary composite achieved a maximum specific capacitance of 928.56 F/g (at 40 mV s⁻¹), surpassing that of individual building blocks (Fig. 7d). Moreover, the SGP composite exhibited 77.68 % capacitance retention after 11,000 cycles (Fig. 7e), underscoring its robust performance and long-term stability.

The exceptional combination of flexible durability, swift electrochemical activity, and robust electrical conductivity exhibited by nanostructured hybrid conductive PANI hydrogels renders them exceptionally suited for flexible energy storage devices, leveraging electrostatic attraction. Das et al. employed in situ polymerization of aniline monomer in conjunction with N,N'-dibenzoyl-L-cystine (D) and MoS₂ QD (M) to fabricate a composite hydrogel (DMP). Fig. 8a provides a structural depiction of the composites interconnected via electrostatic interaction and hydrogen bonding [104]. During aniline polymerization in the presence of D, PANI (P) chains form as emeraldine salt, characterized by positively charged radical cation groups. Consequently, electrostatic interactions between these radical cations and carboxylate anions, coupled with hydrogen bonding between the imino group of PANI and the >C=O group of D's carboxyl acid groups, facilitate the establishment of a cross-linked polymer network for the composite. At a current density of 1 A g⁻¹ the hybrid DMP7 composite gel demonstrates

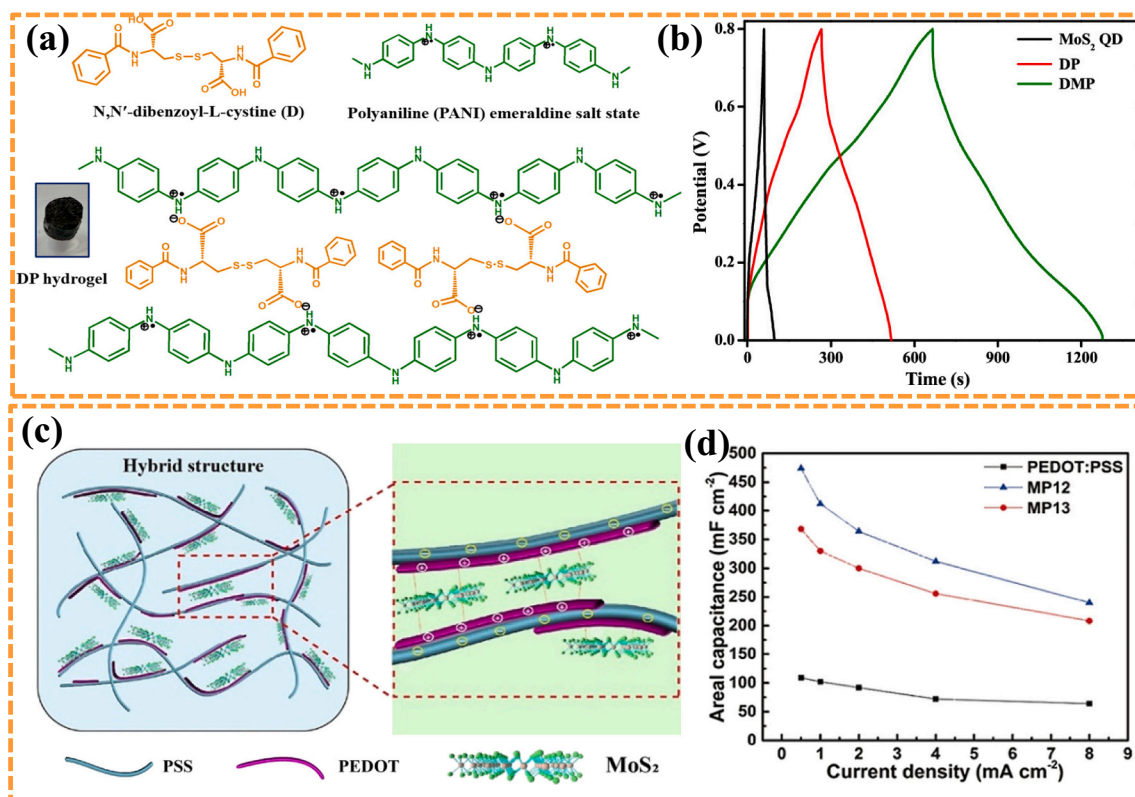


Fig. 8. (a) Structural illustration of gelator (D) and PANI (P), and DP Hydrogel connected with electrostatic interaction and hydrogen bonding. (b) GCD curves of MoS₂ QD, DP and DMP. [104] (c) Schematic diagram of the confined growth of MoS₂ nanosheets driven by hydrogen bonding, cross-linking, and electrostatic attractions. (d) Areal capacitance of PEDOT:PSS, MP12 and MP13 composite at different current density. [109].

a remarkable specific capacitance of 791 F/g, surpassing that of DP (312 F/g) and MoS₂ (51 F/g) with a longer discharge time, as illustrated in Fig. 8b. Furthermore, the interaction between positively charged PEDOT chains and the negatively charged MoS₂ led to the development of a hybrid MoS₂/PEDOT:PSS (MP) hydrogel [109]. This electrostatic attraction not only prevented the agglomeration of MoS₂ nanosheets but also facilitated the physical crosslinking of polymer chains, thereby forming hybrid MP hydrogels (Fig. 8c). Remarkably, the MP12 film demonstrated excellent electrochemical performance in both solid-state PVA-H₃PO₄ electrolyte (360 mF cm⁻²) and aqueous Na₂SO₄ electrolyte (474 mF cm⁻²), better than that of the pure PEDOT:PSS film (109 mF cm⁻²) (Fig. 8d), indicating the enhancement of electrostatic forces in SCs performance.

As evident from the preceding discussion, the combination of CQDs, metal oxides, or MoS₂ nanosheets with CPs is widespread, primarily due to the presence of cationic groups (O or S) within their structure, which can be electrostatically drawn towards the anionic groups in CPs. These interactions lead to alterations in the molecular structure of the composites, preventing agglomeration and resulting in superior specific capacitance and cycle stability compared to pristine CQDs and CPs.

4.2. π - π interaction

The interaction among aromatic groups involving their π bonds is commonly referred to as π - π stacking. The π electron density associated with aromatic rings can generate a quadrupole moment. Typically, the area over the aromatic face of most aromatic rings, like benzene or naphthalene, carries a partial negative charge, while the surrounding area around the ring boundary bears a partial positive charge. However, aromatic rings with strong electron-withdrawing groups tend to pair with other aromatic rings due to the transfer of π electron density from the core rings, creating an electron-deficient central region

characterized by opposite quadrupole moments. [110,111]. For example, the conjugated structure of rGO and aromatic rings of polymers gave rise to the π - π interaction by overlapping each other and the accumulation of massive π - π conjugated electrons will improve the capacity to store charge [112].

As is well-known, carbon materials are commonly composited with CPs through π - π interactions, enhancing electrochemical coupling and subsequently improving the performance of SCs [112]. The chemical bonding between CPs and CNTs or graphene involves strongly conjugated π -bonds in an unsaturated state, allowing for further bonding. These interactions effectively delocalize electrons from the conjugated bonds of the CPs. It's worth noting that a lower number of monomer units per electron transfer results in higher electron transfer capacity for CPs [3].

Relatively weak π - π interactions occur between the PANI quinoid rings and the π -bonds of the graphitic material lattice, facilitating interaction between the PANI matrix and graphitic filler. Consequently, the capacitance and stability of PANI experience significant improvements [113]. For instance, due to the strong attractive force of π - π stacking interaction between aniline monomers and carbon nanohorns (CNHs), polymerization preferentially occurs on CNHs [114]. This interaction within the composites enhances the charge transfer process between the components and promotes effective electron delocalization, leading to favorable SC performance. Consequently, the CNH/graphene/PANI composites demonstrate a capacitance of 765 F/g at 1 A g⁻¹ with 725 F/g retention after 5000 cycles.

The π - π interactions among organic molecules or polymers, meeting specific fundamental criteria, serve as the primary foundation for the noncovalent functionalization of graphene or GO. Wu et al., [115] pioneered the fabrication of a novel continuous 3D structure comprising GO/PPy shell nanospheres. Within this structure, PPy nanoparticles are embedded into GO sheets through in situ polymerization and

interconnected via π - π stacking, hydrogen bonding (O-H...N), and electrostatic interactions (Fig. 9a). The π - π stacking between GO and PPy effectively mitigates the swelling and shrinkage of PPy during the charge-discharge process, indicating a profound alteration of GO on the electronic structure of PPy. Interpreting specific capacitance and energy density values necessitates employing appropriate electrode masses and thicknesses [116]. Significant errors may arise from extremely low discharge rates, especially when paired with small electrode masses. As electrode mass and discharge rate decrease, a larger portion of the signal from faradic reactions, cell leakage current, and other capacitance components within the cell. The PPy/GO composite demonstrates enhanced capacitance, even at a mass loading of 8 mg cm^{-2} , primarily attributed to the smaller particle sizes of PPy nanospheres, which reduce ion diffusion distances and increase the electrode/electrolyte interface for improved ion kinetics. Electrochemical experiments reveal exceptional specific capacitance of 370 F/g (0.5 A g^{-1}) and excellent increased cycle stability, with 91.2 % retention after 4000 cycles, compared to pristine PPy (Fig. 9b). Similarly, a PPy/GO nanowire composite, synthesized via an in-situ chemical polymerization method, features multilayer pyrrole molecules absorbed onto the GO surface, facilitated

by strong π - π interactions, achieving a specific capacitance of 728 F/g at 0.5 A g^{-1} . [117]. Through the loosely packed PPy nanowire structure, the hydrophilic nature of GO may enhance electrode/electrolyte interface regions, facilitating the electrochemical permeability of electrolytes and enabling rapid movement of electrolytic ions within the electrode during charge/discharge operations. Additionally, Tian et al. demonstrated the presence of π - π interactions between the carbon atoms in the hexatomic rings of rGO and PANI. These π - π interactions facilitate the charge transfer process, thereby promoting the delocalization of charge carriers, resulting in superior electrochemical characteristics and higher specific capacitance compared with individual rGO or PANI component [118].

Adhikari et al., [119] demonstrated the effective H- π interaction between graphene and metallocene (ferrocene, Fc), alongside π - π interactions with PANI. Fc serves as a well-known mediator due to its reversibility, regeneration at low potential, and the production of easily controlled stable redox states. The stabilized PANI with Fc@graphene (FcGA) exhibits a resembling and intertwined network. Computational studies confirm the absence of possible interactions between ferrocene and PANI, but with strong π - π interactions between PANI and graphene

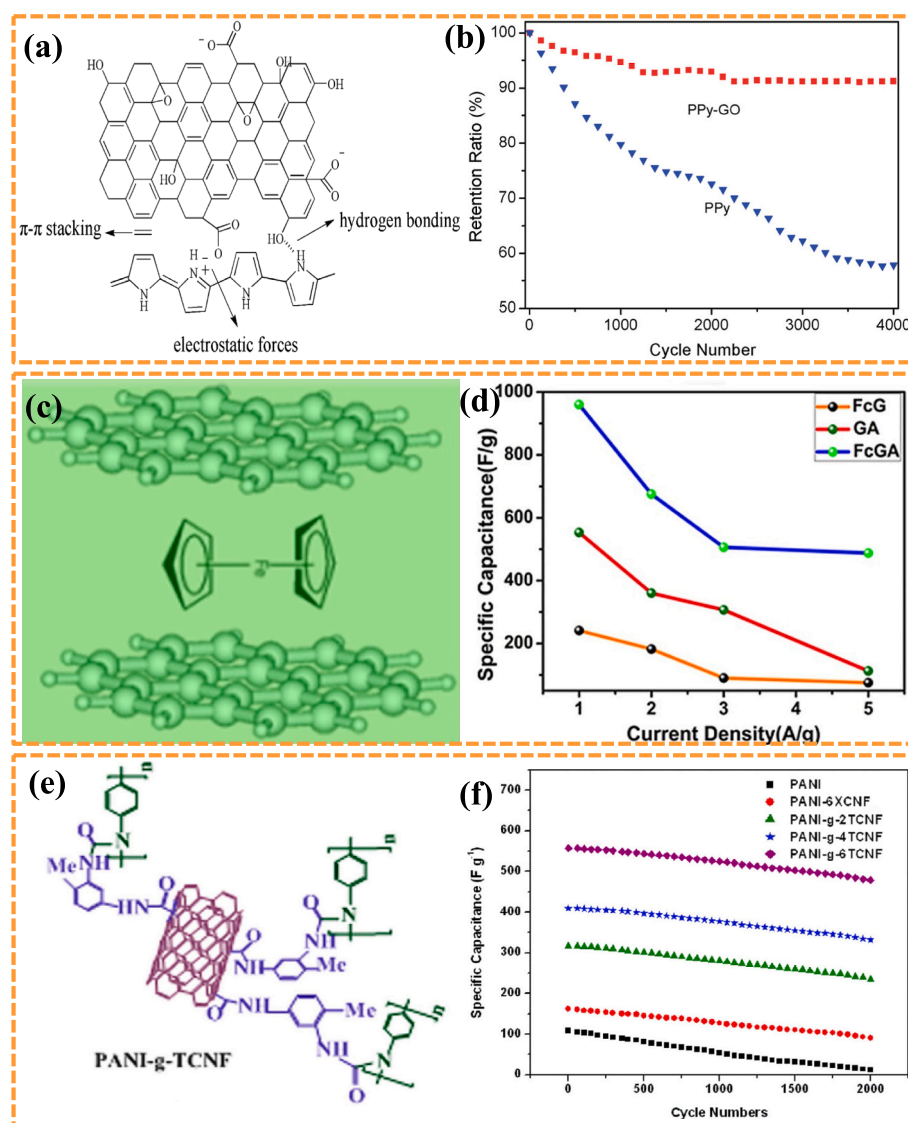


Fig. 9. (a) Proposed possible combining mode of PPy-GO nanocomposites. (b) Cycle stability of PPy and the core-shell PPy-GO electrodes during the long-term charge/discharge process. [115] (c) Sandwiched interaction of FcGA through π - π interactions. (d) Specific capacitance value of FcG, GA and FcGA at different current densities. [119] (e) Schematic diagram for the interaction of PANI and TCNF. (f) Capacitance-retaining performance of the samples PANI, PANI-6XCNF, PANI-g-2TCNF, PANI-g-4TCNF and PANI-g-6TCNF with charge-discharge cycling at 0.3 A g^{-1} . [120].

(Fig. 9c). In addition to the EDLC, the system displays pseudocapacitance owing to the presence of heteroatoms and other functional groups. The PANI-stabilized ternary nanocomposite demonstrates a maximum specific capacitance of 960 F/g (1 A g^{-1}) (Fig. 9d) and remains highly stable, with up to 86 % retention after 5000 cycles, indicating the beneficial impact of the π - π interactions. Except graphene, CNF composites grafted with PANI can be prepared through electrostatic interaction and π - π stacking between graphite surface and aromatic ring in PANI [120]. Here, toluene diisocyanate was primarily functionalized to carboxylate CNF through amidation followed by aniline and polymerized with urea additive. The linkage of phenyl-substituted amide plays an important role between PANI and CNF to develop π - π interactions and facilitates effective faradaic charge transfer through the CNF network (Fig. 9e). The PANI grafting on amidation of CNF (TCNF) facilitates a stronger connection between the two components, leading to uniform growth of PANI chains across the TCNF surface. This enhanced contact effectively reduces the charge-transfer barrier, thereby accelerating charge transfer and potentially improving the electrochemical cycle stability of the composites. Specifically, the composite with 6 % CNF achieves a specific capacitance of 557 F/g with 86 % retention after 2000 cycles at 0.3 A g^{-1} (Fig. 9f).

Additionally, CNTs possess a bucky tube structure with sp^2 hybridized orbitals and a hexagonal mesh structure [121] making them easily composable with CPs via π - π interactions. Typically, CNTs are utilized as additives or templates for synthesizing PPy/carbon composites, enabling the carbon materials to readily interacted with PPy [38,122]. For instance, Wang et al. [123] investigated the electrochemical performance of a composite comprising PPy and SWCNTs, synthesized via the electropolymerization method, which facilitated multilayer adsorption through π - π interactions, enhancing ion transfer from electrolytes to composite films. They achieved a specific capacitance of 200 F/g for PPy/functionalized SWCNT at a scan rate of 200 mV s^{-1} . Additionally, Yang et al. [124] prepared a PPy-based composite with plasma-activated PPy, where the π bonds interaction between CNT and PPy, along with plasma-activated nitrogen and oxygen on the surface, contributed to a specific capacitance of 264 F/g.

The non-covalent interactions, specifically π - π /N-H- π interactions, originating from the MIL-101/PANI (Material Institute Lavoisier-MIL) interface, enhance the charge carrier's mobility (Fig. 10a) [125]. At a current density of 1 A g^{-1} , PANI/MIL-101 exhibits an exceptional specific capacitance of 1197 F/g (178 mF cm^{-2}). Lewis acid-base interactions between imino groups and metal atoms create coordinatively unsaturated metal sites, fostering strong bond complexes. These interactions influence the morphology and improve electrochemical performance by providing a wide surface area in the CV analysis (Fig. 10b). The PANI/MIL-101 composite maintains a specific capacitance of 1197 F/g at 1 A g^{-1} with capacitance retention up to 81 % after 10,000 cycles (Fig. 10c). The interaction between PANI and MIL-101 prompts the PANI molecular chain to infiltrate the pores, thereby enhancing the electrochemical and charge transfer properties of PANI/MIL-101. However, an excessive amount of insulator (MIL-101) may adversely affect electrochemical performance and reduce specific capacitance. Additionally, aniline undergoes in-situ chemical polymerization in the presence of Zirconium 1,4-dicarboxybenzene UiO-66 (University of Oslo) to prepare the PANI/UiO-66 hybrid. Fig. 10d illustrates the interaction and conduction between PANI and UiO-66. At a current density of 1 A g^{-1} , the PANI/UiO-66 achieves an excellent specific capacitance of 647 F/g, maintaining 90 % of its initial value after 5000 cycles of charging and discharging (2.5 A g^{-1}). It is proposed that the enhanced charge-carrier transfer resulting from the inclusion of PANI and the non-covalent π - π /N-H- π interaction between PANI and UiO-66 contributes to the capacitive characteristics [126].

CP hydrogels present a compelling option due to their robust electrical conductivity, rapid ion diffusion, exceptional mechanical properties, and expansive specific surface area, all synergistically complementing hydrogels and organic conductors [127]. At a current

density of 0.5 A g^{-1} , PIn aerogel exhibits an enhanced capacitance of 168 F/g. Moreover, the positively charged PIn readily nucleates on the surfaces of GO sheets, facilitated by the abundance of -COOH and -OH groups at their borders and defects on the basal planes. Upon hydrothermal reduction of GO, stable hydrogels form, owing to reinforced hydrogen bonding between the amine or hydroxyl groups in PIn and the -COOH or -OH groups in graphene, electrostatic interactions between charged amine and COO-, and π - π stacking between the aromatic rings of PIn and rGO. This unique interaction yields binder-free 3D graphene hydrogels with a large surface area, enhancing electron transport, reducing dynamic electrolyte resistance, and facilitating better interactions with electrolyte ions.

Khasim et al. demonstrated a PEDOT:PSS/rGO composite electrode using ethylene glycol, exhibiting a specific capacitance of 174 F/g. They proposed that the robust π - π interaction between rGO and PEDOT:PSS enhanced the conductivity and ion charge carrier mobility [128]. Conversely, Liu and colleagues synthesized PEDOT:PSS/rGO via a hydrothermal method, where the PEDOT:PSS/rGO fibers self-assembled through π - π stacking and hydrophobic interactions. This yielded a capacitance value of 249 F/g, alongside an energy density of 10.68 Wh kg^{-1} at a power density of 81.25 W kg^{-1} [129].

It is evident that π - π interactions between distinct conjugated systems can lead to entirely different stacking configurations. Thus, further research is needed to comprehensively grasp the electron structures of these components. These interactions can be strong by manipulating the electron density of the components via various modification strategies. Despite this gap, the aforementioned studies provide valuable guidance for assembling composites through π - π interactions.

4.3. Confinement effect

The quantum mechanical phenomenon of “quantum confinement” entails restricting particle mobility in at least one dimension, commonly observed in extremely small physical systems, typically at the nanoscale. Particles, such as electrons, experience effects from this limitation that differ from those in larger systems. Besides quantum size, the confinement effect manifests in carbon-based fibers, 3D structures, metal hydroxides, etc., significantly influencing electrochemical performance. Although the confinement effects in CPs-based composites remain insufficiently studied and unconfirmed by theoretical analyses, certain literature, particularly concerning PANI and PPy, does mention the confinement effect in CPs composites. [130,131].

For instance, Yang et al. proposed the utilization of a single *Calotropis gigantea* fibers (CGF) natural hollow fiber to markedly enhance the performance of flexible SCs, achieving a specified capacitance [131]. Scanning electron microscopy (SEM) images confirmed the confinement of hydrophobic pyrrole molecules by the hydrophobic surfaces of the fibers, both internally and externally (Fig. 11a). These molecules were subsequently electropolymerized in situ to yield a conducting PPy polymer electrode, thereby expanding the accessible area of the electrolyte. Fig. 11b illustrates the areal capacitance as a function of current density, with CC-CGF-PPy exhibiting average values of $785.78 \text{ mF cm}^{-2}$ (341.64 F/g or 19.64 F cm^{-3}) at a current density of 4 mA cm^{-2} , with a capacitance retention of 77 % after 2000 cycles. While optimization studies proposed a minimal increase in PPy weight alongside CGF addition, the confinement of PPy on the CGF surfaces enhances pseudocapacitance owing to the increased PPy weight. Conversely, the substantial hollowness of the embedded CGFs results in a significantly expanded electrolyte-accessible area due to the confinement effect.

The carbon-based 3D-GO confined with PANI was synthesized via in-situ polymerization [132]. During charge/discharge operations, the confinement of aniline within the GO architecture facilitates rapid ion and electron transport. The growth of PANI during polymerization is undoubtedly constrained by the interlayer spacing of the graphene basal plane, creating an efficient channel for both electrolyte ions and electrons. At 0.5 A g^{-1} , this composite exhibits an enhanced capacitance of

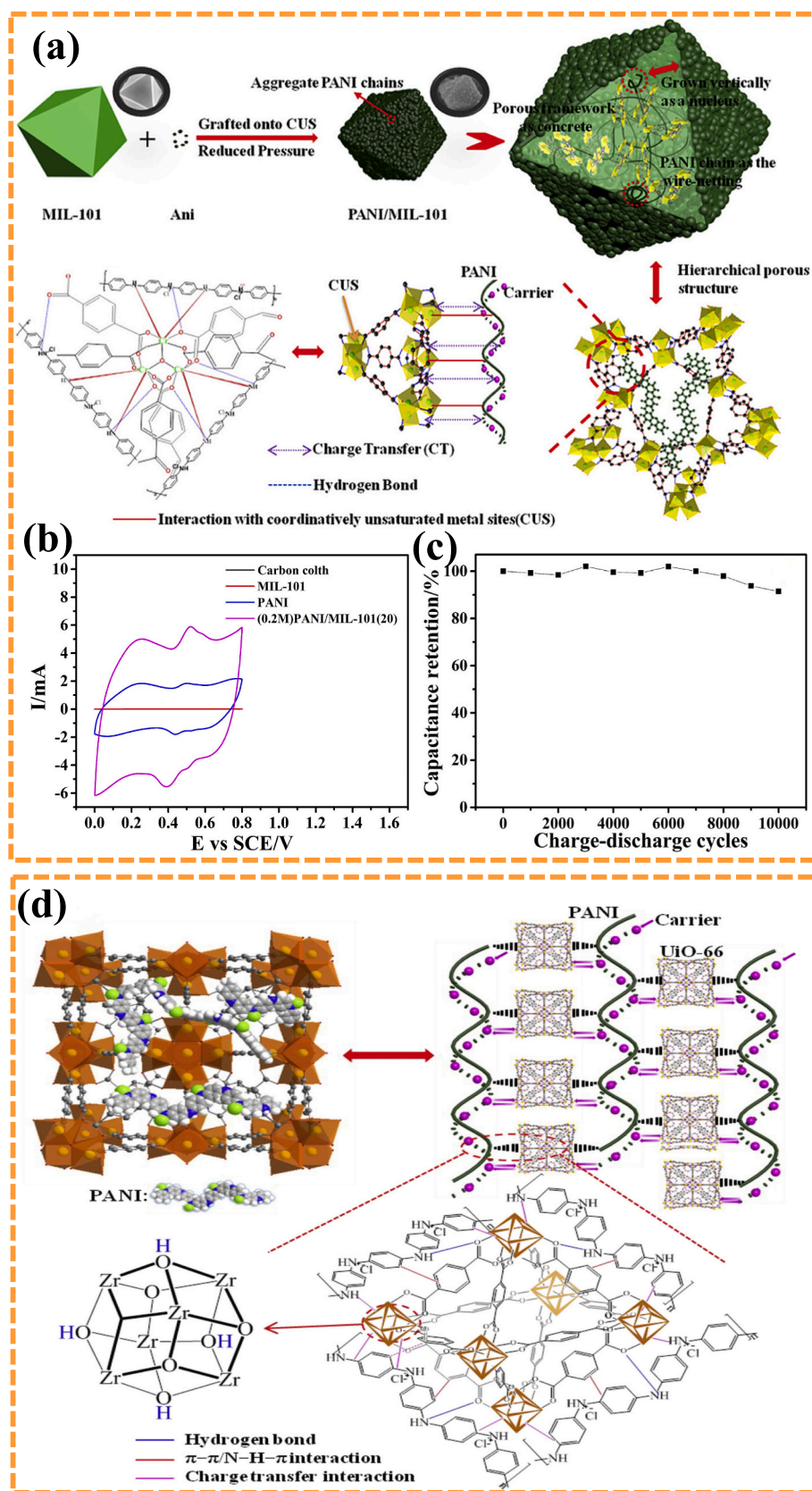


Fig. 10. (a) Schematic illustration of the fabrication process and the interaction between PANI and MIL-101. (b) CV curves of carbon cloth, MIL-101, PANI, (0.2 M) PANI/MIL-101(20) electrodes at 10 mV s⁻¹. (c) Capacitance retention of (0.2 M) PANI/MIL-101(20) during GCD cyclic test at 20 A g⁻¹. [125] (d) Schematic illustration of interaction and carrier conduction between PANI and UiO-66. [126].

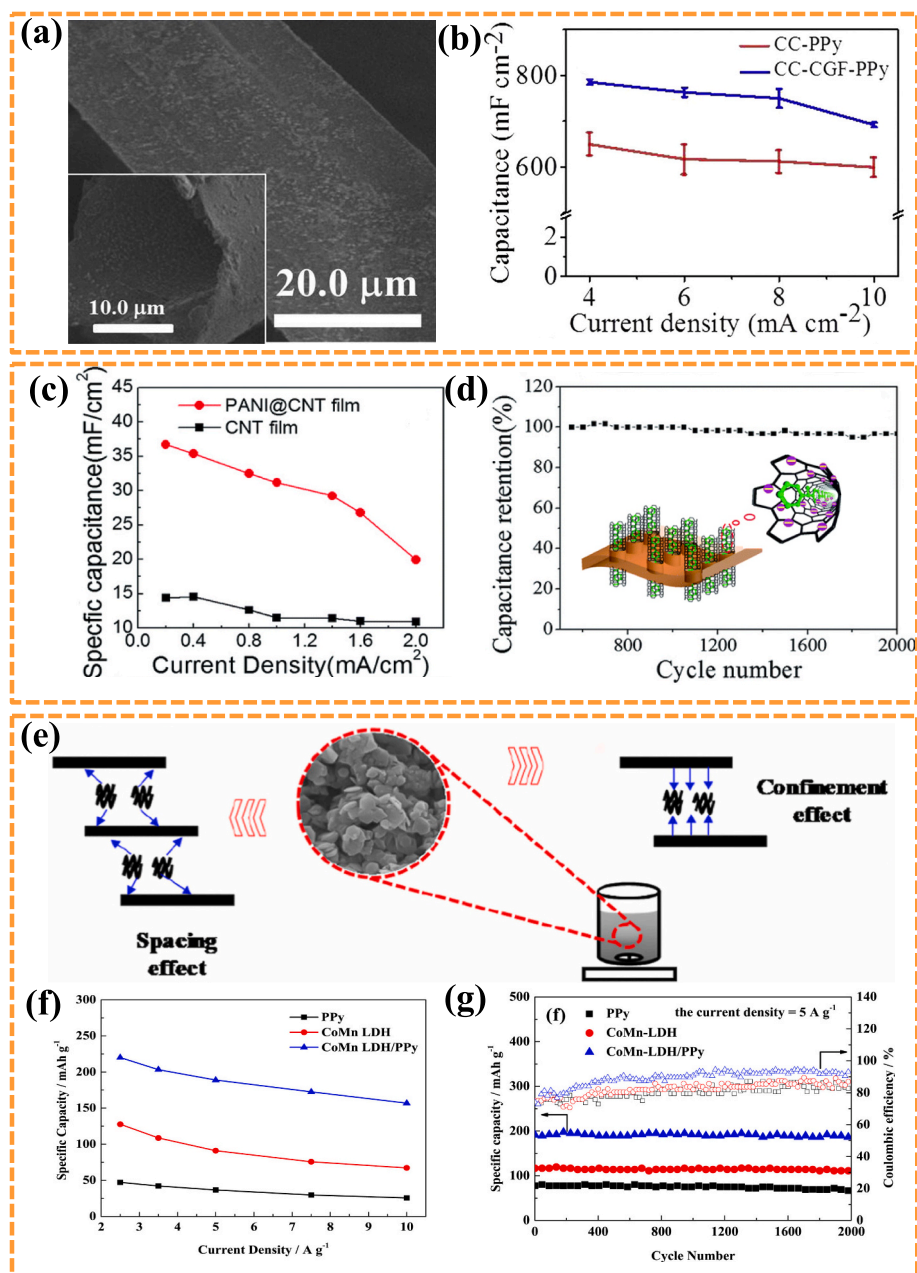


Fig. 11. (a) The SEM images of CGF-PPy and enlarged cross-sectional view (inset). (b) The areal capacitance of the electrodes CC-PPy and CC-CGF-PPy as a function of scan rate, with standard deviation of 3 measurements. [131] (c) Unit area of PANI@CNT film (red) and CNT film (black) versus different current densities in 1 M KOH solution. (d) Galvanostatic charge-discharge cycling test of PANI@CNTs at a current density of 1 mA cm^{-2} in 1 M KOH between 0 V and 0.4 V (inset: the scheme of PANI confined in the CNTs). [133] (e) The scheme of confinement effects in sample CoMn LDH/PPy. (f) The variation in the specific capacity of different samples PPy, CoMn LDH and CoMn LDH/PPy as a function of current density. (g) Cyclic stability tests of different samples PPy, CoMnLDH, CoMnLDH/PPy at 5 A g^{-1} . [134]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

780 F/g, surpassing that of PANI alone (323 F/g). Additionally, Wang's group prepared MWCNT-confined PANI using an in-situ electropolymerization method [133]. The graphitic CNT's aligned nanostructure provides effective charge transfer channels and shields the organic polymers. Importantly, the constrained CNT nano-channels prevent structural alterations in PANI during repeated charge/discharge cycles, extending the device's lifespan. This confined PANI/MWCNT composite demonstrates a notable electrochemical capacitance of 296 F/g at 1.6 A g^{-1} (20 mF cm^{-2} @ 2 mA cm^{-2}) with a capacitance loss of less than 5 % after 2000 cycles (Fig. 11c, d). Hence, CPs encapsulated within MWCNTs show considerable promise for fabricating thin, flexible SC electrodes.

Layered double hydroxides (LDHs) possess stable and versatile characteristics attributable to the adjustable interlayer anions and intralayer metal cations within their structure. The controlled sizes and high theoretical capacities of various Co-based LDHs, such as CoNi LDHs, render them valuable as electrode materials for battery applications. Particularly noteworthy are CoMn LDHs, which are garnering increased attention owing to the superb pseudocapacitive performance of Mn and enhanced conductivity facilitated by the formation of COOH. Nonetheless, the semiconducting nature and tendency for simple stacking or agglomeration of LDHs may lead to low specific capacity and poor rate capability. For instance, cobalt manganese (CoMnLDH)/PPy demonstrates excellent electrochemical performance, attributed to the

synergistic effects of confinement from CoMnLDH layers and spacing provided by conductive PPy particles. [134]. Fig. 11e depicts the confinement effects in CoMnLDH/PPy. According to SEM and Brunauer-Emmett-Teller (BET) analysis, the author demonstrated that the addition of PPy to the composite decreased the stacking of LDH platelets, while the confinement effect of LDH reduced the size of PPy particles (Fig. 11f, g). Electrochemical studies revealed that the composite material exhibited a specific capacitance of 1494 F/g at 2.5 A g⁻¹. The confinement effects enhance the pseudocapacitance of PPy by restricting its particle size, potentially contributing to the observed increase in redox peaks for CoMn LDH/PPy.

From the above discussion about confinement effect, the CPs has crystalline nature embedded in the amorphous nature. The crystalline nature was responsible for the electrical conduction in CPs where inside the island of CPs there was the occurrence of confinement charge carriers. The particle size of composites was the reason towards the good effect of confinement, therefore, micrometre to nano-sized particles will be distributed more homogeneously to form ultrathin CPs electrodes to enhance the electrochemical performances. These findings highlight the significance of the confinement effect in influencing and dictating the behavior of supercapacitance as well as EDLC properties.

4.4. Synergistic effect

Synergy entails the amalgamation of two to three elements, resulting in a collective outcome surpassing that of the individual components [135]. The positive synergistic effect manifests through the mere blending of additives, demonstrating enhanced properties compared to single elements [136]. Similarly, in polymer/carbon or polymer/metal oxides/carbon composites, the synergistic effect consistently arises due to various factors. CPs inherently exhibit conductivity and pseudocapacitance behavior, while carbonaceous materials offer high conductivity, substantial surface area, and EDLC characteristics [137] and metal oxide possesses the behavior of pseudocapacitance [138]. These advantages collectively fortify mechanical and other physicochemical properties, mitigating the limitations of individual components [139].

To improve the energy density of the SCs, compositing of the CPs and metal oxide is widely adopted. Composites such as PANI/zirconium dioxide (ZrO₂) maintain substantial capacitance even after numerous cycles [140]. Similarly, when other metal oxides like SnO₂, cerium oxide (CeO₂), ruthenium oxide (RuO₂), and MnO₂ are incorporated with CPs, the electrical conductivity and charge storage capacitance experience augmentation. Furthermore, combining MXenes or metal organic frameworks (MOFs) with CPs can yield outstanding SCs units, showcasing exceptional cycle stability due to preserved structural integrity and high specific charge capacitance. Subsequently, the ensuing section will delve into a detailed examination of the synergistic effects observed in such composites.

CeO₂ is a rare earth oxide compound known for its oxygen vacancy, pivotal in facilitating outstanding redox electrochemical performances. Its advantageous oxygen capacity arises from the facile redox cycle between Ce³⁺ and Ce⁴⁺, enhancing its electrochemical attributes [141,142]. Introducing a dopant ion with a comparable or smaller radius than Ce⁴⁺ into the CeO₂ lattice host typically engenders lattice defects and surplus oxygen vacancies. Leveraging the exceptional properties of nickel (Ni), the Ni-doped CeO₂ emerges as a compelling choice, given nickel's high electrochemical performance. For instance, the interconnection between conducting PANI and nickel (Ni)-CeO₂ establishes efficient conductive pathways, facilitating enhanced Ni-CeO₂ utilization. This configuration demonstrates a specific capacitance of 866 F/g at 1 A g⁻¹ and remarkable long-term cycling stability (10,000 cycles, 85.6 % capacitance retention), attributed to the synergistic effect [143]. PANI enhances conductivity and contributes to high capacitance, while Ni-CeO₂ stabilizes the PANI matrix and mitigates swelling, shrinking, and cracking during charge-discharge cycles, rendering Ni-CeO₂-PANI a promising candidate for SCs. Similarly, PPy

nanotube/MnO₂ [144], PPy/vanadium oxide (V₂O₅) [145] represent notable PPy-based metal oxide composites, showcasing significant specific capacitance due to synergistic effects. Integration of CNTs as additives or templates for PPy/carbon composite synthesis, covalently bonded with PPy, enhances electrochemical performance, particularly cycle stability, through synergistic effects [38,122]. In the case of RuO₂/PEDOT:PSS hybrid thin films, the cooperative action of PEDOT:PSS and RuO₂ nanoparticles were investigated [146]. The RuO₂/PEDOT:PSS hybrid thin films exhibit areal capacitance of 1.2 mF cm⁻² in solid-state SCs owing to the synergistic effects from the highly conductive PEDOT:PSS and pseudocapacitive RuO₂. Similarly, PPy composites incorporating nanostructured MnO_x displayed 463 F/g, attributed to the synergistic effects between PPy and various redox states of Mn (from Mn to Mn²⁺ and Mn²⁺ to Mn³⁺) metal oxides, alongside the homogeneous dispersion of nanostructured MnO_x [147].

MOFs have attracted considerable attention due to their unique attributes such as high porosity, expansive surface area, nanoscale regularity, and structural versatility [148]. Integration of CPs with MOFs or their derivatives yields enhanced stability, improved charge transport, heightened electrical conductivity, increased porosity, and notable redox characteristics. MOF@CP composites feature active ions within their structure, boasting a superior surface area compared to pure CPs [149]. One such example is the synthesis of Co-MOF/PANI hybrid through an in-situ chemical oxidative polymerization process, as documented by Iqbal et al. The resulting Co-MOF/PANI composite exhibits a pore volume of 0.9811 cm³ g⁻¹, a specific area of 156.59 cm² g⁻¹, and a pore size of 25.06 nm. Notably, the pore size falls within the mesoporous range, which augments electrochemical storage capabilities. These findings underscore the enhanced specific area offered by MOF/PANI composites, facilitating greater charge accumulation and storage capacity. The synergistic interaction between MOF and PANI translates into a significant increase in capacitance, as evidenced by electrochemical studies, revealing a specific capacitance of 504 F/g at a current density of 1 A g⁻¹ [150].

Numerous researchers have extensively investigated PPy/MOF composites, revealing enhanced electrochemical properties attributable to their synergistic effects [151–154]. For instance, NiMOF/PPy composites demonstrated a specific capacitance of 1815 F/g at 1 A g⁻¹, attributed to the synergistic interaction between PPy and the highly stable redox-active Ni-MOF [152]. Additionally, Jiao and colleagues harnessed the catalytic capabilities of MOFs to fabricate a hybrid Zn/Ni-MOF@PPy composite [155]. They elucidated that PPy not only expanded interlayer spacing but also facilitated favorable charge transport, leveraging the high redox properties of Zn/Ni-MOF. In addition to extending the Zn/Ni-MOF's layer gap, the conductive polymer chains offer advantageous charge transport pathways. Zn/Ni-MOFs have a 2D layered structure that allows for transmitting OH⁻ (faradaic reactions). As a result, this energy storage device may utilize the beneficial properties of both the battery and the SCs to concurrently improve the energy and power density. The Zn/Ni-MOF@PPy exhibits exceptional electrochemical performance of 202 F/g at 1 A g⁻¹ (cycle stability of 94.4 % after 3000 cycles), and with varying PPy loads, a “trade-off effect” between ion diffusion kinetics and electrical conductivity was found.

Furthermore, ternary composites comprising CPs, metal oxides, and carbon sources have emerged as a recent area of interest in SCs research. Synergistic effects are prevalent in ternary CP-based composites, crucially contributing to improved electrochemical performance. Moreover, the construction of ternary composites addresses issues such as heavy agglomeration, low surface area, and limited active sites.

Jiang et al. conducted both experimental and theoretical investigations on the synergistic effects of PANI, PPy, and PTh with rGO and titanium oxide (TiO₂) [156]. Their study revealed that the combined effects of these various components significantly enhanced electrochemical performance synergistically. Fig. 12a illustrates the electrochemical performances of G-TiO₂-PTh, G-TiO₂-PPy, and G-TiO₂-PANI.

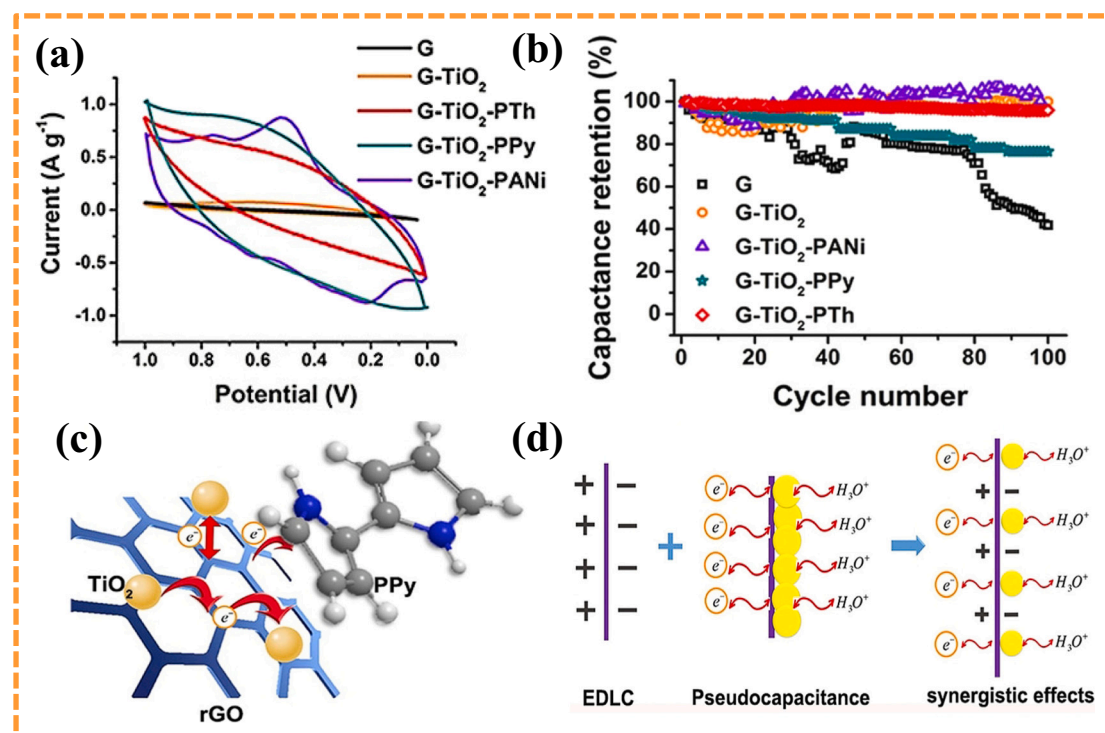


Fig. 12. (a) Comparison of CV curves of G, G-TiO₂, G-TiO₂-PTh, G-TiO₂-PPy, and G-TiO₂-PANI at the scan rate of 5 mV s⁻¹. (b) Capacitance retention cycle stability of G, G-TiO₂, G-TiO₂-PTh, G-TiO₂-PPy and G-TiO₂-PANI. (c) Scheme: synergistic effects of rGO-TiO₂ and different conductive polymers. (d) Scheme: synergistic effects of EDLC and pseudocapacitance behavior. [156].

Additionally, a notable enhancement in cycle stability was observed for G-TiO₂-CPs after the incorporation of TiO₂ (Fig. 12b). To elucidate the improvement mechanism, the authors theoretically predicted and investigated the synergistic effects of the prepared composites. The reversible redox reactions of TiO₂, influencing the pseudocapacitance behavior of PANI, were found to contribute to the synergistic effects. Two mechanisms were proposed to clarify the enhancement of electrochemical performance resulting from synergistic effects: (i) the role of incorporating different additives and (ii) the interaction of physico-chemical properties displayed by the two materials. Regarding the first mechanism, (i) rGO sheets acted as material bridges, facilitating effective electron transfer between TiO₂ and CPs; (ii) rGO exhibited EDLC behavior, enhancing overall electrode electrochemical performance as a carbon-based material; (iii) oxygen-containing groups on rGO sheets provided strong interfacial interactions and electrical connections between rGO and TiO₂, as depicted in Fig. 12c. The second mechanism, depicted in Fig. 12d from the perspective of EDLC and pseudocapacitance, showcased the various contributions of CPs to the electrochemical performance of binary and ternary composite electrode materials. (i) PANI, PPy, and PTh contributed to pseudocapacitance, wherein rapid electrochemical faradaic reactions occurred at the interface between the electrolyte and the CPs. (ii) The sp² hybridized structure of CPs determined a conductive mechanism comparable to that of graphene, ensuring facile ion transmission between the CPs phase and rGO during the electrochemical process. This facilitated quick and efficient ion insertion or extraction, enhancing the electrochemical performances of the composite electrodes [157].

Das et al. synthesized the PANI/MoO₃/graphene nanoplatelets (PMG) composite via in-situ polymerization [158]. The addition of MoO₃ to PANI/graphene nanoplatelets notably increased the active surface area, as evidenced by the expanded surface area of the CV curve. The PMG composite has a large electrode-electrolyte interfacial area available for redox processes during charging and discharging owing to the synergistic contributions of the components through mutual

interactions. Consequently, a specific capacitance of 734 F/g with 92.4 % retention after 1000 cycles was achieved, attributed to the synergistic interactions between ions from PANI and graphene. The high specific capacitance value of the PMG composite can be attributed to the conducting network chain formed by the adhesion of the fiber-shaped PANI over the GNP/MoO₃. The greater specific capacitance of the composite is ascribed to the presence of GNP and PANI, which improves the electrode/electrolyte contact surface area (113.331 m²/g) through critical interactions, promoting electrolytic or electronic ion transport. Additionally, the conducting network chain is formed by the adhesion of the fiber-shaped PANI over the GNP/MoO₃. This chain helps facilitate the intense π - π interactions between the particles. The decreased electronic resistance (11.3 Ω) expresses the PMG excellent conductivity as well as the ease with which the electrolyte diffuses and penetrates the composite. The combined impact of the elements leads to better ion/electron diffusion or transportation at the material-electrolyte contact.

In another study, Chandan Kumar et al. developed a promising boron nitride (BN)/CNT/PANI composite, leveraging the synergistic effects of its constituents [159]. BN has a crucial function in preventing restacking properties, altering surface shape, and forming super-lattices to improve electrochemical performance. This composite exhibited increased capacitance due to the combined contributions of EDLC-type CNT, pseudocapacitive PANI and BN with the capacity to alter surface morphology and active sites. and PANI. Additionally, its mesoporous nature, high porosity, and large surface area further contributed to its remarkable electrochemical performance (387 F/g at 1 A g⁻¹). The presence of inter- and intra-chain electron hopping facilitated efficient electron transport within the ternary composite BN/CNT/PANI.

PPy-based ternary composites represent a significant advancement in achieving long-term stability alongside enhanced electrochemical performance. Combining metal oxides with PPy/carbon has proven highly successful in boosting the performance of PPy-based electrodes. Recently, the Co₃O₄ has emerged as a promising material for SCs, boasting a high theoretical capacitance of 3560 F/g [160]. By

incorporating Co_3O_4 into PPy/rGO, improved performance was achieved, leveraging the synergistic effect between rGO and the pseudocapacitance of Co_3O_4 /PPy [161]. The cycle stability of PPy/rGO/ Co_3O_4 is supported by the faradaic reaction of PPy and H^+ induced by OH^- and Co_3O_4 . This mechanism enhances cyclic stability compared to PPy/carbon composites. Moreover, the synergetic effect between graphene and PPy, stemming from their excellent conductivity and shared sp^2 hybridized structure, facilitates easy and rapid electron transfer between the two materials, thereby increasing long-term cycle stability and overall electrochemical performance.

Kong and co-workers [162] presented a novel approach employing nickel cobalt oxide (NiCo_2O_4) nanowires integrated into carbon textiles with PPy for pseudocapacitors. NiCo_2O_4 , known for its ability to form various nanostructures, features short surfaces and elongated ion diffusion paths coupled with low conductivity. Polymerization could tailor pyrrole monomers into diverse sizes on the carbon textile by adjusting the temperature. Fig. 13a illustrates the charge-discharge mechanism of NiCo_2O_4 nanowires in SCs. The combination of PPy's high conductivity and efficient ion transport pathways, along with NiCo_2O_4 , yields a remarkable specific capacitance of 2244 F/g, with pseudocapacitive behavior evident from the redox peaks of NiCo_2O_4 (M-O/M-OH, M = Ni, Co). The fabricated NiCo_2O_4 @PPy within the carbon textile (Fig. 13b) exhibited impressive cycle stability, retaining 82.9 % after 10,000 cycles, surpassing NiCo_2O_4 nanowires alone. This enhanced capacitance is primarily attributed to the pseudocapacitive properties of both PPy and NiCo_2O_4 , with the latter serving as a robust scaffold for the PPy nanostructures. The pores between the nanowires and nanospheres of NiCo_2O_4 @PPy facilitate electrolyte access during charging and discharging. The mesoporous nanostructure of NiCo_2O_4 @PPy with a high surface area ($97.48 \text{ m}^2/\text{g}$) creates more pathways for electrolyte penetration, allowing electrolyte ions to reach more electro-active locations more efficiently. In another instance, spinel ferrite-based materials were widely used with CPs and influenced the electrochemical performances through their magnetic and electrical properties. The Mn^{2+} and Fe^{3+} ions underwent a reaction in an alkaline

environment under solvothermal conditions to nucleate and develop manganese ferrite (MnFe_2O_4) nanoparticles on the hydrophilic surface of GO. Simultaneously, GO underwent an initial reduction to rGO. Subsequently, to prepare rGO/ MnFe_2O_4 /PPy ternary hybrids, pyrrole monomers were polymerized on the surface of rGO/ MnFe_2O_4 . Here, rGO and PPy acted as a conductive network, enhancing MnFe_2O_4 active sites and overall electrochemical performance. Pseudocapacitive behavior arose from the various oxidation states of MnFe_2O_4 and the redox contributions of PPy. The resulting PPy/ MnFe_2O_4 /rGO composite exhibited 147.2 F/g at 10 mV s^{-1} with a cycle stability of 95 % capacitance retention after 1000 cycles. These impressive electrochemical performances stem from the spinel-type structure of MnFe_2O_4 and graphene, which offer excellent cycling stability and synergistic effects between each component [163].

The PANI/zinc ferrite (ZnFe_2O_4)/GO ternary composite electrode demonstrated a significantly higher area capacitance of 2169.71 mF/g at 1 mA cm^{-2} compared to any individual component, underscoring the synergistic impact of PANI and the additive [164]. Their observation suggests that incorporating ZnFe_2O_4 into PANI electrodes mitigates contact effects, higher surface area ($57 \text{ m}^2/\text{g}$), thereby enhancing ion diffusion and electron transport. Moreover, the addition of ZnFe_2O_4 appears to reduce material resistivity, resulting in a lower charge transfer resistance (R_{ct} , 11Ω) that facilitates rapid charge transfer and enhances the charge storage capability of the ternary electrodes.

MXene possesses functional groups, including $-\text{OH}$, $-\text{O}$, and $-\text{F}$, on its hydrophilic surface, rendering it a favorable electrode material for SCs. However, like other 2D structured materials, as the mass of a single MXene sheet increase, it becomes more prone to stacking, which detrimentally affects the electrode's specific capacitance and impedes the rapid diffusion of electrolyte ions [165]. Currently, the primary approach to mitigate MXene accumulation involves designing the electrode material structure and support matrix within the MXene material. The synergetic contribution from $\alpha\text{-Fe}_2\text{O}_3$, C-MoS₂, and PEDOT:PSS (M/FMP/M) were recently explored in varying proportions by Li and colleagues [166]. Fe_2O_3 , a common transition metal oxide known for its

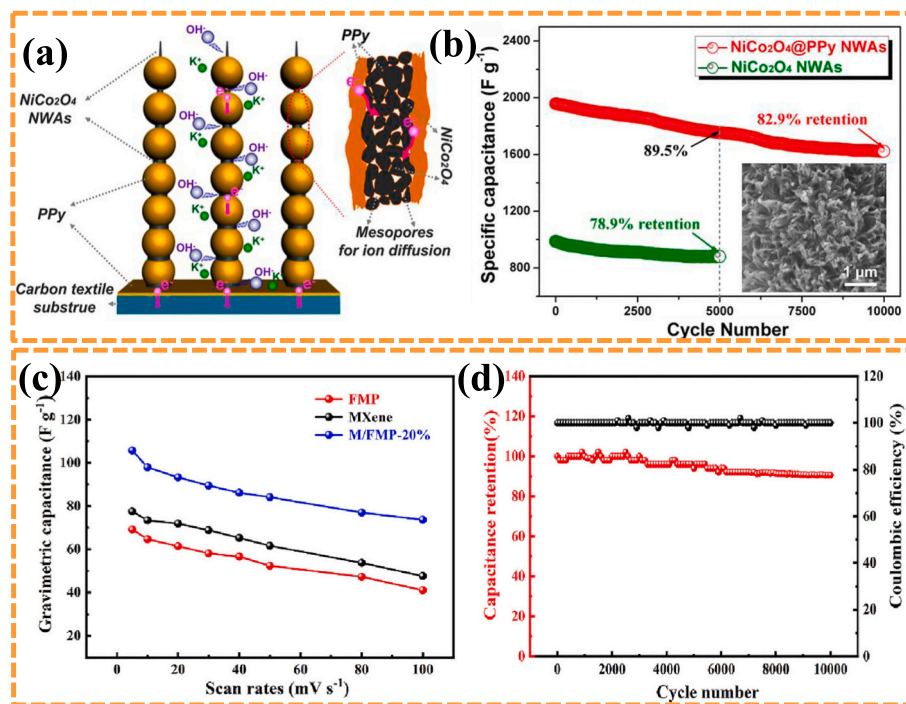


Fig. 13. (a) Schematic representation of rechargeable supercapacitor based on NiCo_2O_4 @PPy NWAs on carbon textiles. (b) Cycling performance of the NiCo_2O_4 @PPy NWAs electrodes (10,000 cycles), compared to NiCo_2O_4 NWAs electrodes (5000 cycles). [162] (c) Gravimetric capacitance of the symmetric supercapacitor-FMP, MXene and M/FMP/M-20 %. (d) Coulombic efficiency and capacitance retention of M/FMP/M-20 % device at 30 mA cm^{-2} with 10,000 cycles. [166].

high redox activity, wide working potential range, affordability, and availability, holds promise as an electrode material for SCs [167]. Conversely, MoS_2 , another commonly used energy storage material, boasts a theoretical specific capacitance of 1504 F/g [168]. However, achieving this theoretical value is challenging due to constraints on conductivity, crystalline phase, and crystallinity. Therefore, combining these materials with PEDOT:PSS helps lower the resistance of $\alpha\text{-Fe}_2\text{O}_3$ and MoS_2 . PEDOT:PSS offers numerous active sites for $\alpha\text{-Fe}_2\text{O}_3$ and C-MoS_2 , enhancing the electrochemical performance of the composite electrode material through synergistic effects. The M/FMP/M-20 composite achieved an areal capacitance of 2700 mF cm^{-2} , while symmetric

SCs composed of M/FMP/M exhibited 528.5 mF cm^{-2} (105.7 F/g) (Fig. 13c). The rate capability of M/FMP/M-20 exhibited a capacitance retention of 71 % after 10,000 cycles at 30 mA cm^{-2} (Fig. 13d). The linked permeable structure of M/FMP/M provides multiple pathways for electrolyte ions, further promoting ion diffusion. This fascinating sandwich electrode arrangement not only increases ion transport but also increases electron transfer. FMP acts as a spacer for MXene, facilitating rapid ion extraction and embedding to maximize MXene pseudocapacitance. Due to its exceptional conductivity and chemical interactions, electrons move swiftly from the current collector to the bottom MXene layer and from MXene to the upper FMP layer.

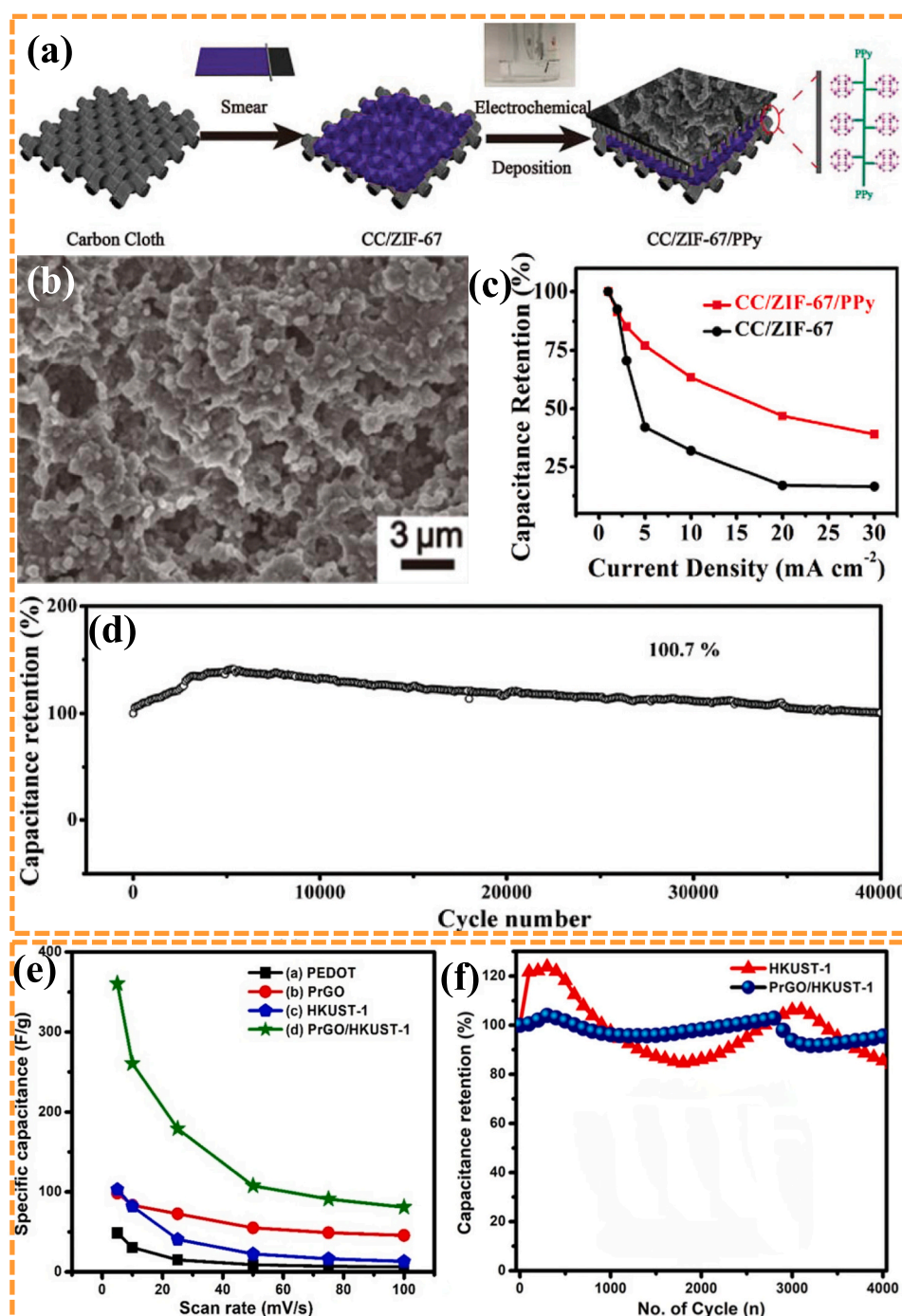


Fig. 14. (a) Schematic illustration for the synthesis of CC/ZIF/PPy composite as a positive electrode material. (b) SEM images of CC/ZIF-67/PPy. (c) The capacitance retention of CC/ZIF-67/PPy and CC/ZIF-67. (d) Cycling performance of ZIF/PPy/10. [169] (e) Specific capacitance values of pure PEDOT, PrGO, HKUST-1 and PrGO/HKUST-1. (f) Capacitance retention of HKUST-1 and PrGO/HKUST-1 up to 4000 cycles. [35].

Based on a conventional MOF ZIF-67 (Zeolitic Imidazole framework) and PPy, Liu developed and created a composite positive electrode material for SCs (Fig. 14a). ZIF-67 and PPy were integrated on carbon cloth using electrochemical deposition technology. Following the electrochemical deposition of pyrrole, ZIF-67 crystals were coated with homogeneous PPy arrays. The conducting relationship between isolated ZIF-67 crystals resembles the nanostick-like morphologies of PPy, creating a porous structural arrangement (Fig. 14b). PPy, serving as a bridge-like pathway, facilitates rapid electron transit to engage with the active sites of ZIF-67. Cobalt in ZIF-67 exists in both Co^{2+} and Co^{3+} states, exhibiting rectangular CV shapes in both pure and composite electrodes. Consequently, the composite electrode consisting of CC, ZIF-67, and PPy (CC/ZIF-67/PPy) demonstrates exceptional capacitance of 180.7 mF cm^{-2} (284.3 F/g) with a capacitance retention ratio of approximately 39.1 % (Fig. 14c). Furthermore, after 40,000 cycles, it achieves very high long-term cycling stability of 100.7 % (Fig. 14d). Among all reported MOF-based SCs electrodes, this one showcases the best performance, leveraging the synergistic effects of PPy adhesion to the interspaces of CC/ZIF-67, formation of a thin film on the surface of ZIF-67, and activation of a large amount of Co^{2+} during charge/discharge to form faradaic pseudocapacitance and EDLC nature [169]. The lower resistance suggests that PPy, acting as a conducting bridge between ZIF-67 particles, not only reduces electronic transfer resistance but also increases ZIF-67's electrochemical charge-transfer rate. Liu et al. fabricated another MOF-based material, NiCoMOF@PNT (where PNTs refer to PPy nanotubes), using the ultrasonic method. The mesopores structure with the specific surface area of $66.5 \text{ m}^2/\text{g}$ increases the surface area and electronic conductivity of NiCo-MOF nanosheets, hence improving their electrochemical performance. NiCoMOF@PNT achieves specific capacitances of 1109 F/g at 0.5 A g^{-1} , indicating favorable rate capacitance. Redox peaks observed in NiCoMOF@PNT denote the involvement of $\text{Ni}^{2+}/\text{Ni}^{3+}$ and $\text{Co}^{2+}/\text{Co}^{3+}$ in faradaic pseudocapacitance reactions [170].

HKUST-1 (Hong Kong University of Science and Technology) a copper-based MOF composed of trimesic acid (organic ligand) and copper ions, exhibits promising characteristics such as pore volume, excellent temperature stability, and a large surface area, making it a potential candidate for SCs applications [171,172]. However, its low electronic conductivity poses a challenge. To address this limitation, carbon materials and CPs are incorporated into HKUST-1 to enhance their mechanical and electrical conductivity properties. HKUST and PEDOT were implemented by Mohandas and associates to achieve adaptable symmetrical SCs [35]. Fig. 14e depicts the correlation between specific capacitance and scan rate, demonstrating that capacitance decreases as the scan rate increases. Higher specific capacitance is achieved at slower scan speeds, allowing electrolyte ions to fully utilize the material's electroactive sites and initiate redox reactions. As an electrode in SCs, PrGO/HKUST film (HKUST-1, rGO, and PEDOT) yields a specific capacitance of 360.5 F/g , maintaining an impressive lifetime of 95 % after 4000 cycles (Fig. 14e, f) due to the synergistic effect of the composite material's EDLC and pseudocapacitance nature. The resistance of PrGO/HKUST-1 before and after the cycle stability test explored 1.16Ω and 1.19Ω , suggesting high intrinsic conductivity. In electrochemical performance, PEDOT exhibits a quasi-rectangular CV shape, indicating pseudocapacitance, while PrGO displays an almost rectangular CV curve characteristic of EDLC. The redox peak on the HKUST-1 CV curve confirms a faradic charge storage mechanism. Notably, the integration of PrGO with HKUST-1 (PrGO/HKUST-1) significantly enhances redox peak currents, primarily driven by HKUST-1's pseudocapacitance feature. The reduction peak signifies the reduction of Cu^{2+} to Cu^+ , while the oxidation peak illustrates the oxidation of Cu^+ to Cu^{2+} . The positive synergetic effect in PrGO/HKUST is enhanced by PEDOT interactions with coordinatively unsaturated metal sites of the MOF, coulombic interactions, π - π stacking, and π interaction with functional groups. Overall, in MOF-based CPs, the hybridization of PANI emerges as a prudent choice for enhancing overall performance, leveraging the

synergistic effects of each material. PANI-based MOF composites exhibit higher capacitance, while PPy/MOF interactions are notably stronger than PANI. Conversely, PEDOT:PSS/MOF composites offer flexible properties and increased conductivity, particularly beneficial for solid-state SCs.

From studies on the synergetic effect, it becomes evident that the good electrochemical performance of these composites can be further enhanced by leveraging the unique properties of each material in synergy. Combining CPs with suitable elements enhances capacitance through synergistic interactions between the constituent building blocks, facilitating the fabrication of efficient SCs for various applications. Among these, ternary CPs-based composites have demonstrated higher specific capacitance. However, further research is required to deepen our understanding of synergetic composites, including theoretical confirmation in future studies.

5. Outlooks and perspective

In this review, we have delved into the interfacial interactions within CPs composites, exploring their preparation methods and electrochemical properties. CPs stand out as highly regarded materials for high energy density and long-life SCs due to their flexibility, affordability, eco-friendliness, structural diversity, and facile functionalization via nanostructured composite engineering. Among these, PANI, PPy, PTH derivatives and Pln are prominent CPs combined with EDLC-based carbon or pseudocapacitive materials, addressing intrinsic limitations such as low capacity, sloping plateau, and poor stability in pure CPs-based SCs through interfacial engineering. These strategies accelerate charge transfer kinetics and mitigate volume expansion, bolstering overall performance.

To fabricate various types of CPs-based composites, non-covalent interactions serve as pivotal binding motifs. Over the past two decades, advancements in understanding fundamental non-covalent interactions have paved the way for developing advanced CPs materials with tunable interactions, crucial for SCs applications. The design principles and preparation techniques discussed herein allow for the fine-tuning of physicochemical, mechanical, morphological, and electrochemical properties in non-covalently bonded CPs composite materials, leveraging hydrogen bonding, electrostatic interactions, and π - π interactions. Despite the significant improvements achieved through non-covalent interactions, the detailed mechanisms within these composites remain elusive, often attributed peripherally to synergistic effects. This merits further exploration through advanced characterization technologies such as synchrotron X-ray absorption spectroscopy and in-situ Raman spectroscopy. Moreover, the reported enhancement mechanisms have yet to be validated through computational techniques and simulations. Computational modelling holds immense potential in unravelling the interfacial interaction mechanisms, guiding the selection of suitable polymer matrices and additives, and optimizing the volume fraction of each phase to enhance composite performance. However, the construction of a reasonable interface model for SCs poses a significant challenge.

Therefore, future endeavors in modelling are anticipated to provide insights into predicting the electrochemical behavior of CPs-based composites, which is essential for the development of high-performance SCs.

Moreover, morphology significantly impacts the electrochemical properties of composites. Confinement emerges as the optimal strategy for tuning CPs composites, achieved by constructing confined carbon/metal-based additives alongside CPs. Additionally, adjusting the additive ratios during CPs composite synthesis is pivotal for homogeneous dispersion within the CPs matrix, effectively reducing interfacial resistance. Furthermore, while many studies have focused solely on the electrochemical properties of CPs composites, it is imperative to also consider factors such as thermal stability, mechanical properties, and processing capabilities to meet practical application requirements.

Finally, the design principles governing the construction of CPs-based composites for SCs adhere to the following key tenets: (1) ensuring compatibility between additives to facilitate composite formation via non-covalent interactions; (2) fine-tuning the morphology and size of the composites through the confinement effect; and (3) leveraging the properties of CPs and additives to mitigate each other's limitations, such as high resistance, sluggish ion transfer, diminished capacitance, or stability degradation due to synergistic effects.

CRedit authorship contribution statement

Berlina Maria Mahimai: Writing – original draft, Conceptualization. **Erling Li:** Writing – review & editing, Validation, Conceptualization. **Jiwei Zhang:** Supervision, Conceptualization. **Jingwei Zhang:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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